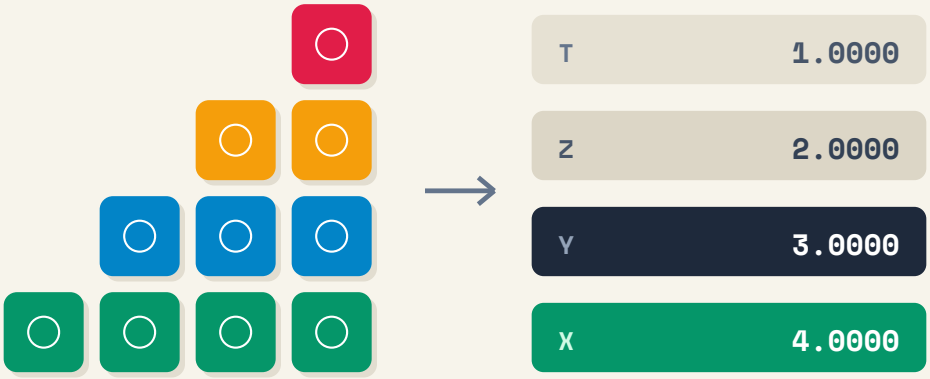
UNIFIED TEACHER'S CONCEPT GUIDE • GRADES K–12

TEACHING WITH THE STACK

Hands-On Mathematics & Concept Facilitation Guide (K–12)



StackCalc

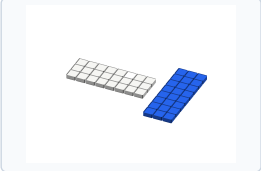
MIDDLE SCHOOL MATH LAB:
10-UNIT TEACHER & PARENT COACH GUIDE

Compact 1-Page Per Unit Solution Keys (Practice A & B #1–#10), Target Misconceptions, Hands-On Block Remediations & Classroom Pacing

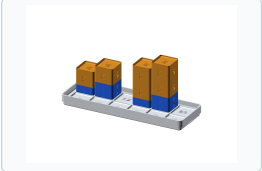
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COMPANION TO FROM BLOCKS TO THE STACK & HANDS-ON CLASSROOM MANIPULATIVES

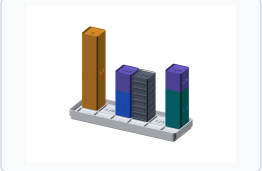
- **Illustrated Companion Book: From Blocks to the Stack: Hands-On Math & the SC-6 Stack Calculator**
- Double-Points Mastery Scoring: Earn +10 points for your 1st Textbook Way and +20 points for verifying with a 2nd Way.
- Hands-On Models: Array Tiles, Prime Factor Towers, Fraction Strips, Place Value Blocks, Two-Tone Counters, Algebra Tiles & Geared Clock.
- **Solve by Hand First, Then Check on SC-6: Every section includes a dedicated SC-6 Calculator Work-Check box with hardware keycaps.**



Counting & Array Tiles



Prime Factor Towers



Fraction Strips



Algebra & Area Tiles



Geared Base-60 Clock

WHAT IS INSIDE THIS EDITION (FRONT MATTER, CORE UNITS & BACK MATTER APPENDICES)

Front Matter (pp. i–ii)

Title Page, Classroom Reproduction License, Pacing Guide & Companion Book Directory.

Core Unit Keys 1–5 (pp. 1–5)

1-Page Coach & Answer Keys for Units 1–5 (Whole Numbers, Fractions/Decimals, Ratios/Percents, Expressions, Area/Time).

Core Unit Keys 6–10 (pp. 6–10)

1-Page Coach & Answer Keys for Units 6–10 (Integers/Coord Plane, Equations/Inequalities, 3D Solids, Statistics, Grade 7 Bridge).

Appendix A (p. 11)

Master Diagnostic Misconception Matrix, Hands-On Block Remediations & 45-Day / 9-Week Pacing Schedule.

Appendix B (p. 12)

Photocopiable Classroom & Homeschool Student Progress Roster & 100-Point Mastery Tower Log.

Why Two Textbook Methods + Hands-On Models + RPN Verification Work

1. THE THREE-PART MASTERY LOOP: SOLVE BY HAND IN TWO WAYS, MODEL, THEN VERIFY

- Step 1 — Solve by Hand in Your 1st Textbook Way (+10 Points): Study the section's Worked Example and solve the problem using your primary textbook method.
- Step 2 — Verify with a 2nd Independent Textbook Way (+20 Points Double): Solve the same problem using a second textbook pathway or hands-on block model.
- Step 3 — Check Your Work on the SC-6 Stack Calculator: Enter the postfix verification sequence shown in the section's SC-6 Check Box to confirm your result.
- Step 4 — Self-Check & Diagnose: Compare your final answer against Appendix E (pp. 107–111) or consult the Unified Teacher's Guide for full setups.
- Mathematical Verification: All 90 worked examples and 900 practice problem solutions are verified against the StackCalc RPNCore engine.
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2. CLASSROOM & HOMESCHOOL REPRODUCTION PERMISSION

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Print Individual Chapters (Units 1–10 Separate PDFs): https://stackcalc.io/curriculum/big-ideas-middle-school-sc6-workbook/

3. THE COMPLETE STACKCALC CURRICULUM LINEUP & COMPANION RESOURCES

BOOK 1 — TEACHING WITH THE STACK: UNIFIED K–12 MASTER TEACHER'S GUIDE (BY CONCEPT)

Organizes all K–12 concept domains and all 10 Middle School Big Ideas units into a single concept-by-concept teaching reference with target misconceptions, hands-on block remediations, and complete worked setups & RPN keys.

BOOK 2 & BOOK 3 — HANDS-ON MATH LAB (K–12) & MIDDLE SCHOOL MATH LAB (BIG IDEAS SUPPLEMENT)

Dedicated lay-flat student workbooks pairing concrete manipulative models, story openers from From Blocks to the Stack and Thinking in the Stack, guided keycap walkthroughs, and compact self-check answer keys.

BOOK 4 — STACK SPEED & FLUENCY: STUDENT SPEED & PROFICIENCY TRIALS WORKBOOK

15 progressive 10-minute timed RPN fluency sprints (Aligned to UIL Calculator Applications & Number Sense), inside-out postfix speed tactics, personal progress graphing, and complete self-check answer keys.

COMPANION TRADE BOOKS & HANDS-ON MANIPULATIVE BLOCK SETS

Read the illustrated stories in From Blocks to the Stack (SC-6 Elementary) and Thinking in the Stack (SC-32 Scientific) on Amazon, and explore physical manipulative kits and free print files at stackcalc.io.

4. FLEXIBLE 45-DAY / FULL-YEAR PACING & ALIGNMENT WITH FROM BLOCKS TO THE STACK

- Daily Classroom Supplement (90 Days): Assign Practice A (Key Ideas) on Day 1 of a textbook section and Practice B (Modeling Real Life) on Day 2.
- 45-Day Catch-Up / Summer Bridge (9 Weeks): Complete 1 Section per day (5 problems in 2 Ways = 100 Points on Practice A, plus 2 Real-Life problems on Practice B).
- Story-First Hook: Begin each of the 10 Units by reading the 1-page Unit Story Opener together and exploring the matching chapter in From Blocks to the Stack.
- Self-Checking Without Guessing: Students write their paper-and-pencil steps first, verify on the SC-6 calculator, and check final answers in Appendix E.

Unit 1: Numerical Expressions & Factors — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 1 • Hands-On Models: Counting & Array Studio (Counting & Array Tiles) + Prime Factor Studio (Prime Factor Towers)

Target Misconception: Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).

Hands-On Block Remediation: Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 1.1 (pp.4-5): WHOLE NUMBER OPERATIONS & DIVISION WITH REMAINDERS

Methods: W1: Standard Long Division | W2: Partial Quotients | W3: Equal-Groups Array Model

PRACTICE A (p.4): KEY IDEAS (#1-#10)

ExA: 5 R 3

A1: 5 R 1
Setup: 31 ÷ 6 (Q & R)
SC-6: 31 [ENTER] 6 [g][÷R]

A2: 9 R 2
Setup: 47 ÷ 5 (Q & R)
SC-6: 47 [ENTER] 5 [g][÷R]

A3: 8 R 2
Setup: 58 ÷ 7 (Q & R)
SC-6: 58 [ENTER] 7 [g][÷R]

A4: 9 R 4
Setup: 85 ÷ 9 (Q & R)
SC-6: 85 [ENTER] 9 [g][÷R]

A5: 12 R 2
Setup: 146 ÷ 12 (Q & R)
SC-6: 146 [ENTER] 12 [g][÷R]

A6: 7 full lines, 5 cm left
Setup: 250 ÷ 35 (Q & R)
SC-6: 250 [ENTER] 35 [g][÷R]

A7: 117
Setup: (14 × 8) + 5
SC-6: 14 [ENTER] 8 [×] 5 [+]

A8: 9 R 3 (Rem < 8)
Setup: 75 ÷ 8 (Q & R)
SC-6: 75 [ENTER] 8 [g][÷R]

A9: 19 rovers, 2 tires left
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

A10: 8 tents (7 full + 1 for the 29th explorer)
Setup: 29 ÷ 4 (Q & R)
SC-6: 29 [ENTER] 4 [g][÷R]

PRACTICE B (p.5): REAL LIFE (#1-#10)

ExB: 17 vans (16 full vans + 1 van with 7 students)

B1: 8 full packs, 5 cells left
Setup: 53 ÷ 6 (Q & R)
SC-6: 53 [ENTER] 6 [g][÷R]

B2: 11 figures, 6 tickets left
Setup: 94 ÷ 8 (Q & R)
SC-6: 94 [ENTER] 8 [g][÷R]

B3: 20 canoes (19 full + 1 with 2 campers)
Setup: 78 ÷ 4 (Q & R)
SC-6: 78 [ENTER] 4 [g][÷R]

B4: 9 full racks, 2 on last
Setup: 110 ÷ 12 (Q & R)
SC-6: 110 [ENTER] 12 [g][÷R]

B5: 28 knights, 2 m left
Setup: 450 ÷ 16 (Q & R)
SC-6: 450 [ENTER] 16 [g][÷R]

B6: 18 full, 19 caches total
Setup: 165 ÷ 9 (Q & R)
SC-6: 165 [ENTER] 9 [g][÷R]

B7: 187 marbles
Setup: (12 × 15) + 7
SC-6: 12 [ENTER] 15 [×] 7 [+]

B8: 8 full trains, 8 on final
Setup: 200 ÷ 24 (Q & R)
SC-6: 200 [ENTER] 24 [g][÷R]

B9: 9 pizzas (5 slices left)
Setup: 67 ÷ 8 (Q & R)
SC-6: 67 [ENTER] 8 [g][÷R]

B10: 11 full crates, 17 apples left
Setup: 325 ÷ 28 (Q & R)
SC-6: 325 [ENTER] 28 [g][÷R]

SEC 1.2 (pp.6-7): POWERS AND EXPONENTS

Methods: W1: Multiply Step-by-Step | W2: Group Factors in Pairs | W3: Square & Cube Array Model

PRACTICE A (p.6): KEY IDEAS (#1-#10)

ExA: 4³ = 16, 4⁴ = 64

A1: 5 × 5 = 25
Setup: 5^2
SC-6: 5 [x²]

A2: 49 (vs. 14)
Setup: 7^2
SC-6: 7 [x²]

A3: 3 × 3 × 3 = 27
Setup: 3^3
SC-6: 3 [x³]

A4: 125
Setup: 5^3
SC-6: 5 [x³]

A5: 32
Setup: 2^5
SC-6: 2 [ENTER] 5 [g][y^x]

A6: 1,296
Setup: 6^4
SC-6: 6 [ENTER] 4 [g][y^x]

A7: 10,000 (4 zeros) and 1
Setup: 10^4
SC-6: 10 [ENTER] 4 [g][y^x]

A8: 16 (2×2×2×2 = 16)
Setup: 2^4
SC-6: 2 [ENTER] 4 [g][y^x]

A9: 96 kN (3 × 32)
Setup: 2^5 × 3
SC-6: 2 [ENTER] 5 [g][y^x] 3 [×]

A10: 8 m³ per room; 64 m³ total
Setup: $2^3 \times 8$
SC-6: 2 [x³] 8 [×]

PRACTICE B (p.7): REAL LIFE (#1-#10)

ExB: 81 shares

B1: 8² = 64 pixels
Setup: 8^2
SC-6: 8 [x²]

B2: 6³ = 216 blocks
Setup: 6^3
SC-6: 6 [x³]

B3: 2^6 = 64 bacteria
Setup: 2^6
SC-6: 2 [ENTER] 6 [g][y^x]

B4: 4 × 4 × 4 × 4 = 256 players
Setup: 4^4
SC-6: 4 [ENTER] 4 [g][y^x]

B5: 144 + 25 = 169 ft²
Setup: $12^2 + 5^2$
SC-6: 12 [x²] 5 [x²] [+]

B6: 5^4 = 625 starlings
Setup: 5^4
SC-6: 5 [ENTER] 4 [g][y^x]

B7: 14 meters
Setup: √(196)
SC-6: 196 [f][√x]

B8: 2^5 = 32 > 5² = 25 (by 7)
Setup: $2^5 - 5^2$
SC-6: 2 [ENTER] 5 [g][y^x] 5 [x²] [-]

B9: 1,024 bytes
Setup: 2^10
SC-6: 2 [ENTER] 10 [g][y^x]

B10: 27 + 343 = 370 in³
Setup: $3^3 + 7^3$
SC-6: 3 [x³] 7 [x³] [+]

SEC 1.3 (pp.8-9): ORDER OF OPERATIONS

Methods: W1: PEMDAS Step-by-Step Funnel | W2: Distributive Property / Fraction-Bar Grouping

PRACTICE A (p.8): KEY IDEAS (#1-#10)

ExA: 72

A1: 11
Setup: 5 + (2 × 3)
SC-6: 5 [ENTER] 2 [ENTER] 3 [×] [+]

A2: 4
Setup: (14 + 6) ÷ (7 - 2)
SC-6: 14 [ENTER] 6 [+] 7 [ENTER] 2 [-] [÷]

A3: 6
Setup: 30 - (4 × 6)
SC-6: 30 [ENTER] 4 [ENTER] 6 [×] [-]

A4: 84
Setup: (9 + 3) × (15 - 8)
SC-6: 9 [ENTER] 3 [+] 15 [ENTER] 8 [-] [×]

A5: 1
Setup: 50 - (3 + 4)²
SC-6: 50 [ENTER] 3 [ENTER] 4 [+] [x²] [-]

A6: 20
Setup: 120 - (4 × (3 + 2)²)
SC-6: 120 [ENTER] 4 [ENTER] 3 [ENTER] 2 [+] [x²] [×] [-]

A7: 2 (24 ÷ 12)
Setup: (48 - (4 × 6)) ÷ (3 + (2 × 4.5))
SC-6: 48 [ENTER] 4 [ENTER] 6 [×] [-] 3 [ENTER] 2 [ENTER] 4.5 [×] [+]..

A8: 24
Setup: (3² + (15 ÷ 3)) - (2 × 4)
SC-6: 3 [x²] 15 [ENTER] 3 [÷] [+] 2 [ENTER] 4 [×] [-]

A9: $92 (60 + 32)
Setup: (5 × 12) + (4 × 8)
SC-6: 5 [ENTER] 12 [×] 4 [ENTER] 8 [×] [+]

A10: $62 (90 - 28)
Setup: (45 × 2) - 28
SC-6: 45 [ENTER] 2 [×] 28 [-]

PRACTICE B (p.9): REAL LIFE (#1-#10)

ExB: 25

B1: 4(9 + 5) = $56
Setup: (9 + 5) × 4
SC-6: 9 [ENTER] 5 [+] 4 [×]

B2: 45 + 4(7) = $73
Setup: 45 + (4 × 7)
SC-6: 45 [ENTER] 4 [ENTER] 7 [×] [+]

B3: 101 points
Setup: (100 - (3 × 8)) + 25
SC-6: 100 [ENTER] 3 [ENTER] 8 [×] [-] 25 [+]

B4: 10³ - 6³ = 64 m³
Setup: $10^3 - 6^3$
SC-6: 10 [x³] 6 [x³] [-]

B5: $73
Setup: (2 × 14) + (5 × 9)
SC-6: 2 [ENTER] 14 [×] 5 [ENTER] 9 [×] [+]

B6: 70 points
Setup: ((3² + 5) × 6) - 14
SC-6: 3 [x²] 5 [+] 6 [×] 14 [-]

B7: 10 bracelets per box
Setup: (72 - 12) ÷ 6
SC-6: 72 [ENTER] 12 [-] 6 [÷]

B8: 42 (÷ and × go left to right)
Setup: ((48 ÷ 6) × 4) + 10
SC-6: 48 [ENTER] 6 [÷] 4 [×] 10 [+]

B9: 12 muffins per basket
Setup: ((5 × 12) + (3 × 8)) ÷ 7
SC-6: 5 [ENTER] 12 [×] 3 [ENTER] 8 [×] [+] 7 [÷]

B10: 74
Setup: ((4³ - 14) ÷ (2 + 3)) + 8²
SC-6: 4 [x²] 14 [-] 2 [ENTER] 3 [+] [÷] 8 [x²] [+]

SEC 1.4 (pp.10-11): PRIME FACTORIZATION

Methods: W1: Factor Tree (Pair A) | W2: Factor Tree (Pair B) | W3: Division Ladder

PRACTICE A (p.10): KEY IDEAS (#1-#10)

ExA: 60 = 2² × 3 × 5

A1: 2² × 3
Setup: Factors(12)
SC-6: 12 [g][FACT]

A2: 2³ × 3²
Setup: Factors(36)
SC-6: 36 [g][FACT]

A3: 29 is prime; 91 is composite (7 × 13)
Setup: Factors(91)
SC-6: 29 [PRIME], 91 [g][FACT]

A4: 3² × 5
Setup: Factors(45)
SC-6: 45 [g][FACT]

A5: 2³ × 3²
Setup: Factors(72)
SC-6: 72 [g][FACT]

A6: 2² × 3 × 7
Setup: Factors(84)
SC-6: 84 [g][FACT]

A7: 2 × 3 × 5 × 7
Setup: Factors(210)
SC-6: 210 [g][FACT]

A8: 2³ × 3² × 7
Setup: Factors(504)
SC-6: 504 [g][FACT]

A9: 7 × 11 × 13
Setup: Factors(1001)
SC-6: 1001 [g][FACT]

A10: 2^4 × 3² × 5²
Setup: Factors(3600)
SC-6: 3600 [g][FACT]

PRACTICE B (p.11): REAL LIFE (#1-#10)

ExB: 2² × 3 × 5 (e.g., 8 × 15 or 10 × 12)

B1: 1×48, 2×24, 3×16, 4×12, 6×8; 2^4 × 3
Setup: Factors(48)
SC-6: 48 [g][FACT]

B2: 2 × 3³
Setup: Factors(54)
SC-6: 54 [g][FACT]

B3: 51 is composite (3 × 17)
Setup: Factors(51)
SC-6: 51 [g][FACT]

B4: 2^4 × 5
Setup: Factors(80)
SC-6: 80 [g][FACT]

B5: 126
Setup: (2 × 3²) × 7
SC-6: 2 [ENTER] 3 [x²] [×] 7 [×]

B6: 2² × 5²
Setup: Factors(100)
SC-6: 100 [g][FACT]

B7: 2² × 5 × 7
Setup: Factors(140)
SC-6: 140 [g][FACT]

B8: 2^4 × 3²
Setup: Factors(144)
SC-6: 144 [g][FACT]

B9: Both give 2 × 3² × 5
Setup: Factors(90)
SC-6: 90 [g][FACT]

B10: 2 × 3 × 5 × 7 = 210
Setup: ((2 × 3) × 5) × 7
SC-6: 2 [ENTER] 3 [×] 5 [×] 7 [×]

SEC 1.5 (pp.12-13): GREATEST COMMON FACTOR (GCF) & LEAST COMMON MULTIPLE (LCM)

Methods: W1: List Factors & Multiples | W2: Prime Factorization | W3: Prime Factor Venn Diagram

PRACTICE A (p.12): KEY IDEAS (#1-#10)

ExA: GCF = 10, LCM = 100

A1: GCF = 6, LCM = 36
Setup: GCF(12, 18); LCM(12, 18)
SC-6: 12 [ENTER] 18 [GCF] / [f][LCM]

A2: GCF = 4, LCM = 24
Setup: GCF(8, 12); LCM(8, 12)
SC-6: 8 [ENTER] 12 [GCF] / [f][LCM]

A3: GCF = 5, LCM = 75
Setup: GCF(15, 25); LCM(15, 25)
SC-6: 15 [ENTER] 25 [GCF] / [f][LCM]

A4: GCF = 7, LCM = 70
Setup: GCF(14, 35); LCM(14, 35)
SC-6: 14 [ENTER] 35 [GCF] / [f][LCM]

A5: 24
Setup: GCF(48, 72)
SC-6: 48 [ENTER] 72 [GCF]

A6: 28
Setup: GCF(84, 140)
SC-6: 84 [ENTER] 140 [GCF]

A7: GCF = 15, LCM = 225
Setup: GCF(45, 75); LCM(45, 75)
SC-6: 45 [ENTER] 75 [GCF] / [f][LCM]

A8: 60 minutes
Setup: LCM(15, 20)
SC-6: 15 [ENTER] 20 [f][LCM]

A9: 180 seconds (3 minutes)
Setup: LCM(45, 60)
SC-6: 45 [ENTER] 60 [f][LCM]

A10: 12 bags (3 red, 4 blue per bag)
Setup: GCF(36, 48)
SC-6: 36 [ENTER] 48 [GCF]

PRACTICE B (p.13): REAL LIFE (#1-#10)

ExB: 6 packs (3 pretzels and 4 fruit bars per pack)

B1: 14 kits (2 bandages, 3 wipes each)
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF]

B2: 60 years
Setup: LCM(12, 15)
SC-6: 12 [ENTER] 15 [f][LCM]

B3: 24 (3 packs of hot dogs, 2 packs of buns)
Setup: LCM(8, 12)
SC-6: 8 [ENTER] 12 [f][LCM]

B4: 16 singers per row
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF]

B5: 10:12 AM (72 minutes later)
Setup: LCM(18, 24)
SC-6: 18 [ENTER] 24 [f][LCM]

B6: 12 inches
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF]

B7: GCF = 7, LCM = 42
Setup: LCM(14, 21)
SC-6: 14 [ENTER] 21 [GCF] 14 [ENTER] 21 [f][LCM]

B8: 60 minutes
Setup: LCM(LCM(4, 6), 10)
SC-6: 4 [ENTER] 6 [f][LCM] 10 [f][LCM]

B9: 15 bouquets (3 gold, 5 silver each)
Setup: GCF(45, 75)
SC-6: 45 [ENTER] 75 [GCF]

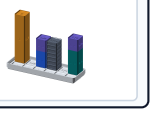
B10: GCF=5, LCM=60; 5×60 = 300 = 15×20
Setup: GCF(15, 20) × LCM(15, 20)
SC-6: 15 [ENTER] 20 [GCF] 15 [ENTER] 20 [f][LCM] [×]

Unit 2: Fractions & Decimals — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 2 • Hands-On Models: Fraction Studio (Fraction Strips) + Place Value Studio (Place Value Blocks)

Target Misconception: Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.

Hands-On Block Remediation: Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 2.1 (pp.15-16): EQUIVALENT FRACTIONS, MIXED NUMBERS & LCD ADDITION

Methods: W1: Divide Once by GCF / Use LCD | W2: Step-by-Step Common Factors | W3: Fraction Strips Model

PRACTICE A (p.15): KEY IDEAS (#1-#10)

ExA: 2/3

A1: 2/3
Setup: Simp(8/12)×2
SC-6: [.] 8 [.] 12 [ENTER] [g][SIMP]×2

A2: 8/3 = 2 2/3 trays
Setup: Simp(48/18)
SC-6: [.] 48 [.] 18 [ENTER] [g][SIMP]

A3: 13/5 and 2.6
Setup: Improper(2 3/5)
SC-6: 2 [.] 3 [.] 5 [ENTER] [f][d/c]

A4: 3 5/6 pizzas (3 whole, 5 sixths)
Setup: 23/6
SC-6: [.] 23 [.] 6 [ENTER]

A5: 5/6
Setup: 1/2 + 1/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 3 [+]

A6: 1 3/20 gallons (23/20)
Setup: 3/4 + 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [+]

A7: 3 5/6 feet
Setup: 1 1/2 + 2 1/3
SC-6: 1 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 3 [+]

A8: 2 7/12 feet
Setup: 5 1/3 - 2 3/4
SC-6: 5 [.] 1 [.] 3 [ENTER] 2 [.] 3 [.] 4 [-]

A9: 7/8 climbed; 1/8 remains
Setup: (1/2 + 1/4) + 1/8
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [+] [.] 1 [.] 8 [+]

A10: 5 1/4 pints
Setup: 3 1/2 + 1 3/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [+]

PRACTICE B (p.16): REAL LIFE (#1-#10)

ExB: 2 11/12 miles (35/12 miles)

B1: 5/8 cup
Setup: 3/8 + 1/4
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 4 [+]

B2: 3/4 complete; 1/4 remains
Setup: Simp(18/24)
SC-6: [.] 18 [.] 24 [g][SIMP]

B3: 13/8 and 1.625 in
Setup: Improper(1 5/8)
SC-6: 1 [.] 5 [.] 8 [ENTER] [f][d/c] [FDISP]

B4: 11/24 foot
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

B5: 3 5/6 pizzas (23/6)
Setup: 2 1/3 + 1 1/2
SC-6: 2 [.] 1 [.] 3 [ENTER] 1 [.] 1 [.] 2 [+]

B6: 2 3/4 feet (11/4)
Setup: 4 1/2 - 1 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 1 [.] 3 [.] 4 [-]

B7: 3/5 = 0.6
Setup: Simp(15/25)
SC-6: [.] 15 [.] 25 [g][SIMP] [FDISP]

B8: 1/5 liter (0.2 L)
Setup: (2/5 + 3/10) - 1/2
SC-6: [.] 2 [.] 5 [ENTER] [.] 3 [.] 10 [+] [.] 1 [.] 2 [-]

B9: 4 5/12 yd used; 7/12 yd left
Setup: 2 2/3 + 1 3/4
SC-6: 2 [.] 2 [.] 3 [ENTER] 1 [.] 3 [.] 4 [+]

B10: 4 5/8 miles
Setup: (3/8 + 2 1/2) + 1 3/4
SC-6: [.] 3 [.] 8 [ENTER] 2 [.] 1 [.] 2 [+] 1 [.] 3 [.] 4 [+]

SEC 2.2 (pp.17-18): MULTIPLYING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply Straight Across | W2: Cross-Cancel Common Factors First | W3: Area Model / Distributive Split

PRACTICE A (p.17): KEY IDEAS (#1-#10)

ExA: 3/8

A1: 1/3
Setup: 1/2 × 2/3
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×]

A2: 1/2
Setup: 2/3 × 3/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 4 [×]

A3: 1/3
Setup: 3/5 × 5/9
SC-6: [.] 3 [.] 5 [ENTER] [.] 5 [.] 9 [×]

A4: 6 cups
Setup: 16 × 3/8
SC-6: 16 [ENTER] [.] 3 [.] 8 [×]

A5: 4
Setup: 1 3/4 × 2 2/7
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 2 [.] 7 [×]

A6: 3
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

A7: 9 cups
Setup: 2 1/4 × 4
SC-6: 2 [.] 1 [.] 4 [ENTER] 4 [×]

A8: 3 1/3 (Less than 5 because 2/3 < 1)
Setup: 5 × 2/3
SC-6: 5 [ENTER] [.] 2 [.] 3 [×]

A9: 14 7/8 miles (119/8)
Setup: 3 1/2 × 4 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [.] 1 [.] 4 [×]

A10: 1/4 of the input speed
Setup: (1/2 × 2/3) × 3/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 2 [.] 3 [×] [.] 3 [.] 4 [×]

PRACTICE B (p.18): REAL LIFE (#1-#10)

ExB: 4 cups

B1: 2/5 of the garden
Setup: 2/3 × 3/5
SC-6: [.] 2 [.] 3 [ENTER] [.] 3 [.] 5 [×]

B2: 2/5 of a full battery
Setup: 4/5 × 1/2
SC-6: [.] 4 [.] 5 [ENTER] [.] 1 [.] 2 [×]

B3: 4 3/8 cups (35/8)
Setup: 1 3/4 × 2 1/2
SC-6: 1 [.] 3 [.] 4 [ENTER] 2 [.] 1 [.] 2 [×]

B4: 3 in²
Setup: 2 1/4 × 1 1/3
SC-6: 2 [.] 1 [.] 4 [ENTER] 1 [.] 1 [.] 3 [×]

B5: 4 miles
Setup: 3 1/5 × 1 1/4
SC-6: 3 [.] 1 [.] 5 [ENTER] 1 [.] 1 [.] 4 [×]

B6: 2/3 yd²
Setup: 3/4 × 8/9
SC-6: [.] 3 [.] 4 [ENTER] [.] 8 [.] 9 [×]

B7: 10 gallons
Setup: (24 × 5/8) × 2/3
SC-6: 24 [ENTER] [.] 5 [.] 8 [×] [.] 2 [.] 3 [×]

B8: 1/8 in²
Setup: 1/2⁸
SC-6: [.] 1 [.] 2 [x^]

B9: 8 ft²
Setup: 3 1/2 × 2 2/7
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 7 [×]

B10: 1/12 of the day (2 hours)
Setup: 1/3 × 1/4
SC-6: [.] 1 [.] 3 [ENTER] [.] 1 [.] 4 [×]

SEC 2.3 (pp.19-20): DIVIDING FRACTIONS AND MIXED NUMBERS

Methods: W1: Multiply by the Reciprocal | W2: Common-Denominator Division | W3: Fraction Strip / Number-Line Model

PRACTICE A (p.19): KEY IDEAS (#1-#10)

ExA: 6

A1: 2
Setup: 1/2 ÷ 1/4
SC-6: [.] 1 [.] 2 [ENTER] [.] 1 [.] 4 [÷]

A2: 4
Setup: 2/3 ÷ 1/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 6 [÷]

A3: 2
Setup: 5/6 × 1/(5/12)
SC-6: [.] 5 [.] 6 [ENTER] [.] 5 [.] 12 [f][1/x] [×]

A4: 8
Setup: 6 ÷ 3/4
SC-6: 6 [ENTER] [.] 3 [.] 4 [÷]

A5: 1 7/8 (15/8)
Setup: 3/4 ÷ 2/5
SC-6: [.] 3 [.] 4 [ENTER] [.] 2 [.] 5 [÷]

A6: 7/8 cup
Setup: 3 1/2 ÷ 4
SC-6: 3 [.] 1 [.] 2 [ENTER] 4 [÷]

A7: 2 2/3 bike frames
Setup: 2/3 ÷ 1/4
SC-6: [.] 2 [.] 3 [ENTER] [.] 1 [.] 4 [÷]

A8: 3
Setup: 5 1/4 ÷ 1 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] 1 [.] 3 [.] 4 [÷]

A9: 5 full slats (5 7/37 -> 3 1/2 in left)
Setup: 96 ÷ 18 1/2
SC-6: 96 [ENTER] 18 [.] 1 [.] 2 [÷]

A10: 6 glasses
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

PRACTICE B (p.20): REAL LIFE (#1-#10)

ExB: 6 shelves

B1: 6 slices
Setup: 3/4 ÷ 1/8
SC-6: [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷]

B2: 12 pebbles
Setup: 4 1/2 ÷ 3/8
SC-6: 4 [.] 1 [.] 2 [ENTER] [.] 3 [.] 8 [÷]

B3: 6 runners
Setup: 3 3/4 ÷ 5/8
SC-6: 3 [.] 3 [.] 4 [ENTER] [.] 5 [.] 8 [÷]

B4: 7 segments
Setup: 5 1/4 ÷ 3/4
SC-6: 5 [.] 1 [.] 4 [ENTER] [.] 3 [.] 4 [÷]

B5: 40 layers
Setup: 6 ÷ 3/20
SC-6: 6 [ENTER] [.] 3 [.] 20 [÷]

B6: 6 sections
Setup: 7 1/2 ÷ 1 1/4
SC-6: 7 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [÷]

B7: 2 meters
Setup: 4 1/2 ÷ 2 1/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [÷]

B8: 8 cups
Setup: 2 2/3 ÷ 1/3
SC-6: 2 [.] 2 [.] 3 [ENTER] [.] 1 [.] 3 [÷]

B9: 5/12 (12/5 × 5/12 = 1)
Setup: 1/(2 2/5)
SC-6: 2 [.] 2 [.] 5 [f][1/x]

B10: 12 pieces
Setup: (10 1/2 - 1 1/2) ÷ 3/4
SC-6: 10 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 2 [-] [.] 3 [.] 4 [÷]

SEC 2.4 (pp.21-22): ADDING, SUBTRACTING, AND MULTIPLYING DECIMALS

Methods: W1: Vertical Place-Value Algorithm | W2: Convert to Fractions/Hundredths | W3: Place-Value Split / Area Model

PRACTICE A (p.21): KEY IDEAS (#1-#10)

ExA: 18.45

A1: 345.67 mm
Setup: (345 + 0.6) + 0.07
SC-6: 345 [ENTER] 0.6 [+] 0.07 [+]

A2: 17.55
Setup: 12.8 + 4.75
SC-6: 12.8 [ENTER] 4.75 [+]

A3: 11.35
Setup: 20 - 8.65
SC-6: 20 [ENTER] 8.65 [-]

A4: 2.7
Setup: 4.5 × 0.6
SC-6: 4.5 [ENTER] 0.6 [×]

A5: 18.74
Setup: (25.4 - 9.86) + 3.2
SC-6: 25.4 [ENTER] 9.86 [-] 3.2 [+]

A6: 8.1 (Estimate: 4 × 2 = 8)
Setup: 3.6 × 2.25
SC-6: 3.6 [ENTER] 2.25 [×]

A7: 0.06 (3/50)
Setup: 0.15 × 0.4
SC-6: 0.15 [ENTER] 0.4 [×] [FDISP]

A8: 35.79 g (35.7864)
Setup: 28.86 × 1.24
SC-6: 28.86 [ENTER] 1.24 [×]

A9: $6.65
Setup: 20 - (8.75 + 4.60)
SC-6: 20 [ENTER] 8.75 [ENTER] 4.60 [+] [-]

A10: 161 meters
Setup: ((30 × 1.20) + (40 × 1.35)) + (50 × 1.42)
SC-6: 30 [ENTER] 1.20 [×] 40 [ENTER] 1.35 [×] [+] 50 [ENTER] 1.42 [×..

PRACTICE B (p.22): REAL LIFE (#1-#10)

ExB: $8.10

B1: $41.24
Setup: 18.75 + 22.49
SC-6: 18.75 [ENTER] 22.49 [+]

B2: $6.35
Setup: 20 - 13.65
SC-6: 20 [ENTER] 13.65 [-]

B3: 0.55 seconds faster
Setup: 13.42 - 12.87
SC-6: 13.42 [ENTER] 12.87 [-]

B4: $7.80
Setup: 2.40 × 3.25
SC-6: 2.40 [ENTER] 3.25 [×]

B5: 1.19 cm²
Setup: 1.4 × 0.85
SC-6: 1.4 [ENTER] 0.85 [×]

B6: $41.40
Setup: 11.5 × 3.60
SC-6: 11.5 [ENTER] 3.60 [×]

B7: 30.35 meters
Setup: (24.8 - 6.35) + 11.9
SC-6: 24.8 [ENTER] 6.35 [-] 11.9 [+]

B8: $35.75
Setup: (2 × 14.50) + 6.75
SC-6: 2 [ENTER] 14.50 [×] 6.75 [+]

B9: 2.5 mm gap
Setup: 170.5 - (4 × 42)
SC-6: 170.5 [ENTER] 4 [ENTER] 42 [×] [-]

B10: $35.00 profit
Setup: (3.20 - 1.80) × 25
SC-6: 3.20 [ENTER] 1.80 [-] 25 [×]

SEC 2.5 (pp.23-24): DIVIDING WHOLE NUMBERS AND DECIMALS

Methods: W1: Power-of-10 Decimal Shift & Long Division | W2: Equivalent Fraction Scaling | W3: Reciprocal Fraction Division

PRACTICE A (p.23): KEY IDEAS (#1-#10)

ExA: 5 R 3 = 5 3/4 = 5.75

A1: 4 R 7, 4 7/8, and 4.875
Setup: 39 ÷ 8
SC-6: 39 [ENTER] 8 [÷] [FDISP]

A2: 2.3
Setup: 18.4 ÷ 8
SC-6: 18.4 [ENTER] 8 [÷]

A3: 45 quarters
Setup: 11.25 ÷ 0.25
SC-6: 11.25 [ENTER] 0.25 [÷]

A4: 8
Setup: 7.2 ÷ 0.9
SC-6: 7.2 [ENTER] 0.9 [÷]

A5: 130
Setup: 15.6 ÷ 0.12
SC-6: 15.6 [ENTER] 0.12 [÷]

A6: 6.3 (6 3/10)
Setup: 9.45 ÷ 1.5
SC-6: 9.45 [ENTER] 1.5 [÷] [FDISP]

A7: $20.80
Setup: 124.80 ÷ 6
SC-6: 124.80 [ENTER] 6 [÷]

A8: 24 (24 × 0.2 = 4.8)
Setup: 4.8 ÷ 0.2
SC-6: 4.8 [ENTER] 0.2 [÷]

A9: 6 laps
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

A10: $1.25 per bottle; 16 bottles
Setup: 7.50 ÷ 6; 20 ÷ 1.25
SC-6: 7.50 [ENTER] 6 [÷] -> 1.25; 20 [ENTER] 1.25 [÷]

PRACTICE B (p.24): REAL LIFE (#1-#10)

ExB: 27 quarters

B1: 4 R 3 = 4 3/8 = 35/8 = 4.375
Setup: 35 ÷ 8
SC-6: 35 [ENTER] 8 [÷] [FDISP]

B2: 16 quarters
Setup: 28 ÷ 1.75
SC-6: 28 [ENTER] 1.75 [÷]

B3: $8.65 each
Setup: 34.60 ÷ 4
SC-6: 34.60 [ENTER] 4 [÷]

B4: 72 layers
Setup: 14.4 ÷ 0.20
SC-6: 14.4 [ENTER] 0.20 [÷]

B5: 15 flowers
Setup: 5.25 ÷ 0.35
SC-6: 5.25 [ENTER] 0.35 [÷]

B6: 6 miles per hour
Setup: 4.5 ÷ 0.75
SC-6: 4.5 [ENTER] 0.75 [÷]

B7: 18 sections
Setup: 22.5 ÷ 1.25
SC-6: 22.5 [ENTER] 1.25 [÷]

B8: 30 cups
Setup: 13.5 ÷ 0.45
SC-6: 13.5 [ENTER] 0.45 [÷]

B9: 7 tickets
Setup: 87.50 ÷ 12.50
SC-6: 87.50 [ENTER] 12.50 [÷]

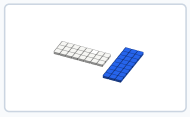
B10: 6
Setup: 10.5 ÷ 1.75
SC-6: 10.5 [ENTER] 1.75 [÷]

Unit 3: Ratios, Rates & Percents — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Fraction Studio (Fraction Strips)

Target Misconception: Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.

Hands-On Block Remediation: Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 3.1 (pp.26-27): RATIOS AND TAPE DIAGRAMS

Methods: W1: Tape Diagram (Equal Boxes) | W2: Equivalent Ratio Fraction | W3: Divide by GCF

PRACTICE A (p.26): KEY IDEAS (#1-#10)

ExA: Part-to-Part = 3:2; Part-to-Whole = 3:5

A1: 2 : 3
Setup: Ratio(12:18)
SC-6: 12 [ENTER] 18 [g][a:b]

A2: 3 : 2
Setup: Ratio(24:16)
SC-6: 24 [ENTER] 16 [g][a:b]

A3: 2 : 5
Setup: Ratio(4:10)
SC-6: 4 [ENTER] 10 [g][a:b]

A4: 18 and 24 (Total = 42)
Setup: 4 × 6
SC-6: 3 [ENTER] 6 [×], 4 [ENTER] 6 [×]

A5: 4 L concentrate, 30 L water
Setup: 34 ÷ 17
SC-6: 34 [ENTER] 17 [÷] -> 2/part

A6: 20 dogs
Setup: (36 ÷ 9) × 5
SC-6: 36 [ENTER] 9 [÷] 5 [×]

A7: 25 seventh graders
Setup: (15 ÷ 3) × 5
SC-6: 15 [ENTER] 3 [÷] 5 [×]

A8: 30 electric guitars (12 acoustic)
Setup: (18 ÷ 3) × 5
SC-6: 18 [ENTER] 3 [÷] 5 [×]

A9: 48 gears and 36 gears
Setup: 84 ÷ 7
SC-6: 84 [ENTER] 7 [÷] -> 12/box

A10: 77 coins total (49 gold, 28 silver)
Setup: (21 ÷ 3) × 11
SC-6: 21 [ENTER] 3 [÷] 11 [×]

PRACTICE B (p.27): REAL LIFE (#1-#10)

ExB: 20 wins (and 12 losses)

B1: Part-to-Part = 3:2; Part-to-Whole = 3:5
Setup: Ratio(15:10)
SC-6: 15 [ENTER] 10 [g][a:b]

B2: 16 strawberries, 12 bananas
Setup: (28 ÷ 7) × 4
SC-6: 28 [ENTER] 7 [÷] 4 [×]

B3: 6 cups yellow (21 cups total)
Setup: (15 ÷ 5) × 2
SC-6: 15 [ENTER] 5 [÷] 2 [×]

B4: 35 secret and 25 open
Setup: (60 ÷ 12) × 7
SC-6: 60 [ENTER] 12 [÷] 7 [×]

B5: 42 gold coins, 18 silver coins
Setup: (24 ÷ 4) × 7
SC-6: 24 [ENTER] 4 [÷] 7 [×]

B6: 3 : 1
Setup: Ratio(48:16)
SC-6: 48 [ENTER] 16 [g][a:b]

B7: 4 : 9 (16 out of 36)
Setup: Ratio(16:36)
SC-6: 16 [ENTER] 36 [g][a:b]

B8: 16 : 9
Setup: Ratio(1920:1080)
SC-6: 1920 [ENTER] 1080 [g][a:b]

B9: 150g almonds, 100g cashews, 250g raisins
Setup: 500 ÷ 10
SC-6: 500 [ENTER] 10 [÷]

B10: 3 : 4
Setup: Ratio(24:32)
SC-6: 24 [ENTER] 32 [g][a:b]

SEC 3.2 (pp.28-29): USING RATIO TABLES AND DOUBLE NUMBER LINES

Methods: W1: Multiplicative Scale Factor | W2: Double Number Line / Repeated Addition | W3: Scale Down Then Up

PRACTICE A (p.28): KEY IDEAS (#1-#10)

ExA: 40 meters

A1: 4:9, 8:18, 12:27, 16:36
Setup: Repeat(+4), Repeat(+9)
SC-6: 4 [g][REP] [+], 9 [g][REP] [+]

A2: 63
Setup: (28 ÷ 4) × 9
SC-6: 28 [ENTER] 4 [÷] 9 [×]

A3: 25
Setup: (60 ÷ 12) × 5
SC-6: 60 [ENTER] 12 [÷] 5 [×]

A4: 60 pages
Setup: (8 ÷ 2) × 15
SC-6: 8 [ENTER] 2 [÷] 15 [×]

A5: 48 ounces
Setup: (18 ÷ 3) × 8
SC-6: 18 [ENTER] 3 [÷] 8 [×]

A6: $15
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

A7: 84 beats
Setup: (60 ÷ 15) × 21
SC-6: 60 [ENTER] 15 [÷] 21 [×]

A8: Ratios scale by multiplication (e.g., ×3 gives 9:15), not ad..
Setup: Ratio(7:9)
SC-6: 7 [ENTER] 9 [g][a:b]

A9: 20 cups
Setup: (12 ÷ 3) × 5
SC-6: 12 [ENTER] 3 [÷] 5 [×]

A10: 40 miles
Setup: (14 ÷ 7) × 20
SC-6: 14 [ENTER] 7 [÷] 20 [×]

PRACTICE B (p.29): REAL LIFE (#1-#10)

ExB: $75

B1: 45 grams
Setup: 15 × 3
SC-6: 15 [ENTER] 3 [×]

B2: 56 scoops
Setup: (16 ÷ 2) × 7
SC-6: 16 [ENTER] 2 [÷] 7 [×]

B3: $15 for 10 notebooks
Setup: (12 ÷ 4) × 5
SC-6: 12 [ENTER] 4 [÷] 5 [×]

B4: 112 beats per minute
Setup: 28 × 4
SC-6: 28 [ENTER] 4 [×]

B5: 260 gallons of sap
Setup: 40 × 6.5
SC-6: 40 [ENTER] 6.5 [×]

B6: 25 pounds
Setup: 150 ÷ 6
SC-6: 150 [ENTER] 6 [÷]

B7: 15 cups of flour
Setup: (6 ÷ 2) × 5
SC-6: 6 [ENTER] 2 [÷] 5 [×]

B8: 75 minutes (1 hr 15 min)
Setup: (90 ÷ 18) × 15
SC-6: 90 [ENTER] 18 [÷] 15 [×]

B9: 8 inches
Setup: (60 ÷ 15) × 2
SC-6: 60 [ENTER] 15 [÷] 2 [×]

B10: 30 lemons
Setup: (9 ÷ 3) × 10
SC-6: 9 [ENTER] 3 [÷] 10 [×]

SEC 3.3 (pp.30-31): RATES, UNIT RATES, AND COMPARING RATES

Methods: W1: Divide to Find the Unit Rate (1 Unit) | W2: Common Multiple Comparison | W3: Double Number Line

PRACTICE A (p.30): KEY IDEAS (#1-#10)

ExA: 63 words per minute

A1: 65 mph
Setup: 195 ÷ 3
SC-6: 195 [ENTER] 3 [÷]

A2: $9 per hour
Setup: 36 ÷ 4
SC-6: 36 [ENTER] 4 [÷]

A3: $1.30 per pound
Setup: 7.80 ÷ 6
SC-6: 7.80 [ENTER] 6 [÷]

A4: 15 grams per hour
Setup: 90 ÷ 6
SC-6: 90 [ENTER] 6 [÷]

A5: Supplier B ($0.17/oz vs. $0.18/oz)
Setup: 3.24 ÷ 18; 4.08 ÷ 24
SC-6: 3.24 [ENTER] 18 [÷]; 4.08 [ENTER] 24 [÷]

A6: 10-pack ($0.82/pen vs. $0.85/pen)
Setup: 5.10 ÷ 6; 8.20 ÷ 10
SC-6: 5.10 [ENTER] 6 [÷]; 8.20 [ENTER] 10 [÷]

A7: 300 miles
Setup: (210 ÷ 3.5) × 5
SC-6: 210 [ENTER] 3.5 [÷] 5 [×]

A8: Pump A (9 gal/min vs. 8 gal/min)
Setup: 45 ÷ 5; 64 ÷ 8
SC-6: 45 [ENTER] 5 [÷]; 64 [ENTER] 8 [÷]

A9: 180 mm
Setup: (45 ÷ 15) × 60
SC-6: 45 [ENTER] 15 [÷] 60 [×]

A10: Car B by 1 mpg (29 mpg vs. 28 mpg)
Setup: 168 ÷ 6; 232 ÷ 8
SC-6: 168 [ENTER] 6 [÷]; 232 [ENTER] 8 [÷]

PRACTICE B (p.31): REAL LIFE (#1-#10)

ExB: Pack B ($1.40 per battery vs. $1.45 per battery)

B1: 150 miles per hour
Setup: 450 ÷ 3
SC-6: 450 [ENTER] 3 [÷]

B2: Box B ($0.70/bar vs. $0.75/bar)
Setup: 7.00 ÷ 10
SC-6: 4.50 [ENTER] 6 [÷] 7.00 [ENTER] 10 [÷]

B3: Spool B ($1.95 vs. $2.20 per 100g)
Setup: 19.50 ÷ 10
SC-6: 11 [ENTER] 5 [÷] 19.50 [ENTER] 10 [÷]

B4: 15 gal/min; 375 gallons
Setup: (180 ÷ 12) × 25
SC-6: 180 [ENTER] 12 [÷] 25 [×]

B5: B (57 taps/min vs. A's 55 taps/min)
Setup: 285 ÷ 5
SC-6: 220 [ENTER] 4 [÷] 285 [ENTER] 5 [÷]

B6: Car A (28 mpg vs. 27 mpg)
Setup: 405 ÷ 15
SC-6: 336 [ENTER] 12 [÷] 405 [ENTER] 15 [÷]

B7: $9 per dog -> 11 dogs
Setup: 99 ÷ (54 ÷ 6)
SC-6: 54 [ENTER] 6 [÷] 99 [x swap y] [÷]

B8: 12-pack ($0.42 vs. $0.45 each)
Setup: 5.04 ÷ 12
SC-6: 3.60 [ENTER] 8 [÷] 5.04 [ENTER] 12 [÷]

B9: 4 inches per hour
Setup: 14 ÷ 3.5
SC-6: 14 [ENTER] 3.5 [÷]

B10: 4.5 m/s -> 90 meters
Setup: (27 ÷ 6) × 20
SC-6: 27 [ENTER] 6 [÷] 20 [×]

SEC 3.4 (pp.32-33): PERCENTS, FRACTIONS, AND DECIMALS

Methods: W1: Equivalent Fraction over 100 | W2: Divide Numerator ÷ Denominator | W3: Percent Bar / Benchmark Model

PRACTICE A (p.32): KEY IDEAS (#1-#10)

ExA: 0.35 and 35%

A1: 0.6 and 60%
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [ENTER] [FDISP]

A2: 0.36 and 36%
Setup: 9/25 -> Dec/Frac
SC-6: [.] 9 [.] 25 [ENTER] [FDISP]

A3: 0.45 and 9/20
Setup: 45 ÷ 100
SC-6: 45 [ENTER] 100 [÷] [FDISP]

A4: 12.5% and 1/8
Setup: 0.125 -> Dec/Frac
SC-6: 0.125 [ENTER] [FDISP]

A5: 270 seats
Setup: 60% of 450
SC-6: 450 [ENTER] 60 [%]

A6: 3,510 grams
Setup: 65% of 5400
SC-6: 5400 [ENTER] 65 [%]

A7: 84%
Setup: 42 ÷ 50
SC-6: 42 [ENTER] 50 [÷]

A8: 1.5 and 1 1/2
Setup: 150 ÷ 100
SC-6: 150 [ENTER] 100 [÷] [FDISP]

A9: 60%, 0.61, 5/8
Setup: 5/8 -> Dec/Frac
SC-6: [.] 5 [.] 8 [FDISP] -> 0.625

A10: 1,400 mAh
Setup: 35% of 4000
SC-6: 4000 [ENTER] 35 [%]

PRACTICE B (p.33): REAL LIFE (#1-#10)

ExB: 75% (3/4 = 0.75)

B1: 0.85 = 85%
Setup: 17/20 × 100
SC-6: [.] 17 [.] 20 [FDISP] 100 [×]

B2: 0.36 = 36%
Setup: 9/25 × 100
SC-6: [.] 9 [.] 25 [FDISP] 100 [×]

B3: 20 + 40 = 60 points
Setup: 40% of 50 + 80% of 50
SC-6: 50 [ENTER] 40 [%] 50 [ENTER] 80 [%] [+]

B4: 104 students
Setup: 65% of 160
SC-6: 160 [ENTER] 65 [%]

B5: 0.15 and 3/20
Setup: Simp(15/100)
SC-6: [.] 15 [.] 100 [g][SIMP]

B6: 45 students total
Setup: 18 ÷ 0.40
SC-6: 18 [ENTER] 0.40 [÷]

B7: $9.00
Setup: 15% of 60
SC-6: 60 [ENTER] 15 [%]

B8: 7/8 = 0.875 = 87.5%
Setup: 21 ÷ 24
SC-6: 21 [ENTER] 24 [÷] [FDISP]

B9: $480 goal
Setup: 360 ÷ 0.75
SC-6: 360 [ENTER] 0.75 [÷]

B10: 58% < 3/5 (60%) < 0.62 (62%)
Setup: 3/5 -> Dec/Frac
SC-6: [.] 3 [.] 5 [FDISP]

SEC 3.5 (pp.34-35): SOLVING PERCENT PROBLEMS, DISCOUNTS & CONVERSIONS

Methods: W1: Find Discount & Subtract | W2: Multiply by Remaining % | W3: Percent Proportion / Ratio Bar

PRACTICE A (p.34): KEY IDEAS (#1-#10)

ExA: Discount = $15; Sale Price = $45

A1: Discount $17; Sale price $68
Setup: 20% of 85
SC-6: 85 [ENTER] 20 [%] [-]

A2: Tax $10.20; Total $130.20
Setup: 8.5% of 120
SC-6: 120 [ENTER] 8.5 [%] [+]

A3: +400% ((75-15)÷15 = 400%)
Setup: %Chg(15->75)
SC-6: 15 [ENTER] 75 [g][%CHG]

A4: 25% decrease
Setup: %Chg(80->60)
SC-6: 80 [ENTER] 60 [g][%CHG]

A5: 25.4 cm
Setup: 10 (to cm)
SC-6: 10 [f][->cm]

A6: 20 in
Setup: 50.8 (to in)
SC-6: 50.8 [g][->in]

A7: $39.20
Setup: 30% of 56
SC-6: 56 [ENTER] 30 [%] [-]

A8: 72
Setup: 18 ÷ 0.25
SC-6: 18 [ENTER] 0.25 [÷]

A9: $144 ($200 -> $160 -> $144)
Setup: 10% of 20% of 200
SC-6: 200 [ENTER] 20 [%] [-] 10 [%] [-]

A10: 81.28 cm and 2.27 kg
Setup: 32 (to cm); 5 (to kg)
SC-6: 32 [f][->cm]; 5 [f][->kg]

PRACTICE B (p.35): REAL LIFE (#1-#10)

ExB: 30% increase; 81.28 cm

B1: Discount = $15; Sale Price = $60
Setup: 20% of 75
SC-6: 75 [ENTER] 20 [%] [-]

B2: Tip = $9.60; Total = $73.60
Setup: 15% of 64
SC-6: 64 [ENTER] 15 [%] [+]

B3: Tax = $3.60; Total = $51.60
Setup: 7.5% of 48
SC-6: 48 [ENTER] 7.5 [%] [+]

B4: 30% increase
Setup: %Chg(40->52)
SC-6: 40 [ENTER] 52 [g][%CHG]

B5: 30% decrease
Setup: %Chg(90->63)
SC-6: 90 [ENTER] 63 [g][%CHG]

B6: 16.09 km
Setup: 10 × 1.609344
SC-6: 10 [ENTER] 1.609344 [×]

B7: 44.09 lb
Setup: 20 (to lb)
SC-6: 20 [g][->lb]

B8: 3.79 liters; 128 fl oz
Setup: 1 (to L)
SC-6: 1 [f][->L]

B9: $72 (10% is taken off $80, not $100)
Setup: 10% of 20% of 100
SC-6: 100 [ENTER] 20 [%] [-] 10 [%] [-]

B10: 68°F
Setup: 20 (to °F)
SC-6: 20 [g][->°F]

Unit 4: Algebraic Expressions & Properties — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Hands-On Models: Algebra Boxes & Factoring Stage (Algebra & Area Tiles) + Counting Tiles (Counting & Array Tiles)

Target Misconception: Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^8, and don't see why $3(x + 4) = 3x + 12$.

Hands-On Block Remediation: Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x^2 is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 4.1 (pp.37-38): WRITING AND EVALUATING ALGEBRAIC EXPRESSIONS

Methods: W1: Substitute in Parentheses & Follow PEMDAS | W2: Equal-Groups Bar / Algebra Tile Model

PRACTICE A (p.37): KEY IDEAS (#1-#10)

ExA: 27

A1: 33
Setup: $(6 \times 7) - 9$
SC-6: 6 [ENTER] 7 [×] 9 [-]

A2: 54
Setup: $(5 \times 8) + 14$
SC-6: 5 [ENTER] 8 [×] 14 [+]

A3: 13
Setup: $40 - (3 \times 9)$
SC-6: 40 [ENTER] 3 [ENTER] 9 [×] [-]

A4: 47
Setup: $6^2 + 11$
SC-6: 6 [x²] 11 [+]

A5: 27
Setup: $(4^2 \times 2) - 5$
SC-6: 4 [x²] 2 [×] 5 [-]

A6: 55
Setup: $(7 \times 4) + (3 \times 9)$
SC-6: 7 [ENTER] 4 [×] 3 [ENTER] 9 [×] [+]

A7: 8
Setup: $(19 + 13) \div 4$
SC-6: 19 [ENTER] 13 [+] 4 [÷]

A8: 15.1
Setup: $(5 \times 3.5) - 2.4$
SC-6: 5 [ENTER] 3.5 [×] 2.4 [-]

A9: 8 + 4.50h; $26
Setup: $(4.50 \times 4) + 8$
SC-6: 4.50 [ENTER] 4 [×] 8 [+]

A10: 150 cm²
Setup: $5^2 \times 6$
SC-6: 5 [x²] 6 [×]

PRACTICE B (p.38): REAL LIFE (#1-#10)

ExB: 99 points

B1: 5g + 6; $26
Setup: $(5 \times 4) + 6$
SC-6: 5 [ENTER] 4 [×] 6 [+]

B2: 50 - 4r; 14 tickets
Setup: $50 - (4 \times 9)$
SC-6: 50 [ENTER] 4 [ENTER] 9 [×] [-]

B3: $225
Setup: $(5^2 \times 8) + 25$
SC-6: 5 [x²] 8 [×] 25 [+]

B4: 82 points
Setup: $(10 \times 7) + (3 \times 4)$
SC-6: 10 [ENTER] 7 [×] 3 [ENTER] 4 [×] [+]

B5: 144 feet
Setup: $3^2 \times 16$
SC-6: 3 [x²] 16 [×]

B6: 96 in²
Setup: $4^2 \times 6$
SC-6: 4 [x²] 6 [×]

B7: 7p + 4d; $41
Setup: $(7 \times 3) + (4 \times 5)$
SC-6: 7 [ENTER] 3 [×] 4 [ENTER] 5 [×] [+]

B8: 78
Setup: $A = \frac{1}{2}((11 + 15) \times 6)$
SC-6: 11 [ENTER] 15 [+] 6 [g][TRI]

B9: 111 minutes (1 hr 51 min)
Setup: $(12 \times 8) + 15$
SC-6: 12 [ENTER] 8 [×] 15 [+]

B10: 128 - 75 = 53
Setup: $(4^3 \times 2) - (5^2 \times 3)$
SC-6: 4 [x²] 2 [×] 5 [x²] 3 [×] [-]

SEC 4.2 (pp.39-40): SIMPLIFYING EXPRESSIONS BY COMBINING LIKE TERMS

Methods: W1: Reorder & Add Coefficients (Properties) | W2: Sort by Tile Shape / Column Alignment

PRACTICE A (p.39): KEY IDEAS (#1-#10)

ExA: 6x + 8

A1: 9x + 13
Setup: $7 + 2$; $4 + 9$
SC-6: 7 [ENTER] 2 [+]; 4 [ENTER] 9 [+]

A2: 5x + 7
Setup: 8 - 3; 11 - 4
SC-6: 8 [ENTER] 3 [-]; 11 [ENTER] 4 [-]

A3: 8y + 6
Setup: 9 - 1
SC-6: 9 [ENTER] 1 [-]

A4: 8a + 13
Setup: 12 - 4; $5 + 8$
SC-6: 12 [ENTER] 4 [-]; 5 [ENTER] 8 [+]

A5: 6x² + 8x + 7
Setup: 4+2=6; 3+5=8
SC-6: Mental / Table Check

A6: 4n² + 5n + 10
Setup: 6-2=4; 9-4=5
SC-6: Mental / Table Check

A7: 5.3x + 2.7
Setup: $3.5 + 1.8$; 4.2 - 1.5
SC-6: 3.5 [ENTER] 1.8 [+]; 4.2 [ENTER] 1.5 [-]

A8: 53 != 80; 5x + 3 is already in simplest form
Setup: 5(10)+3=53 vs 8(10)=80
SC-6: Mental / Table Check

A9: 6x + 16; 76 meters
Setup: 2+2+1+1=6x; 3+3+5+5=16
SC-6: Mental / Table Check

A10: 9x + 8; 53 cm
Setup: 3+2+4=9x; 2+7-1=8
SC-6: Mental / Table Check

PRACTICE B (p.40): REAL LIFE (#1-#10)

ExB: 5x² + 4x + 6

B1: 9x + 9
Setup: $(3 + 1) + 5$
SC-6: 2 [ENTER] 3 [+] 4 [+] 3 [ENTER] 1 [+] 5 [+]

B2: 8x + 12
Setup: $(4 + 2) \times 2$
SC-6: 3 [ENTER] 1 [+] 2 [×] 4 [ENTER] 2 [+] 2 [×]

B3: 4x + 14
Setup: $9 + 5$
SC-6: 6 [ENTER] 2 [-] 9 [ENTER] 5 [+]

B4: 6x² + 4x + 11
Setup: $3 + 8$
SC-6: 4 [ENTER] 2 [+] 7 [ENTER] 3 [-] 3 [ENTER] 8 [+]

B5: 13c + 31
Setup: $12 + 19$
SC-6: 8 [ENTER] 5 [+] 12 [ENTER] 19 [+]

B6: 7x + 19 (coeff = 7, constant = 19)
Setup: $14 + 5$
SC-6: 9 [ENTER] 2 [-] 14 [ENTER] 5 [+]

B7: (1/2)n + 13
Setup: $4 + 9$
SC-6: [.] 3 [.] 8 [ENTER] [.] 1 [.] 8 [+] 4 [ENTER] 9 [+]

B8: 4y + 3.5
Setup: 5.8 - 2.3
SC-6: 2.4 [ENTER] 1.6 [+] 5.8 [ENTER] 2.3 [-]

B9: 6m + 9 (both equal 39 at m=5)
Setup: $((10 - 4) \times 5) + (15 - 6)$
SC-6: 10 [ENTER] 4 [-] 5 [×] 15 [ENTER] 6 [-] [+]

B10: 3x² + 7
Setup: 10 - 3
SC-6: 5 [ENTER] 2 [-] 10 [ENTER] 3 [-]

SEC 4.3 (pp.41-42): THE DISTRIBUTIVE PROPERTY

Methods: W1: Multiply Each Term Inside a(b + c) = ab + ac | W2: Split-Rectangle Area Model | W3: Repeated Addition of Rows

PRACTICE A (p.41): KEY IDEAS (#1-#10)

ExA: 84

A1: 140 + 21 = 161
Setup: $(7 \times 20) + (7 \times 3)$
SC-6: 7 [ENTER] 20 [×] 7 [ENTER] 3 [×] [+]

A2: 4x + 20
Setup: 4×5
SC-6: 4 [ENTER] 5 [×]

A3: 18x + 24
Setup: 6×3=18; 6×4=24
SC-6: Mental / Table Check

A4: 16y - 40
Setup: 8×2=16; 8×5=40
SC-6: Mental / Table Check

A5: 387
Setup: $(9 \times 40) + (9 \times 3)$
SC-6: 9 [ENTER] 40 [×] 9 [ENTER] 3 [×] [+]

A6: 17x + 6
Setup: 3×4+5=17; 3×2=6
SC-6: Mental / Table Check

A7: 12n + 13
Setup: 4×3=12; 20-7=13
SC-6: Mental / Table Check

A8: 5x + 30 (She forgot to multiply 6 by 5)
Setup: 5×6
SC-6: 5 [ENTER] 6 [×]

A9: 14x + 15 (Both equal 43 when x = 2)
Setup: 5×2+4=14; 5×3=15
SC-6: Mental / Table Check

A10: 6(x + 3.50) = 6x + 21; $60 total
Setup: 6×3.50
SC-6: 6 [ENTER] 3.50 [×] -> 6x+21

PRACTICE B (p.42): REAL LIFE (#1-#10)

ExB: 14x + 15

B1: 8(50 + 3) = 400 + 24 = $424
Setup: $(8 \times 50) + (8 \times 3)$
SC-6: 8 [ENTER] 50 [×] 8 [ENTER] 3 [×] [+]

B2: 6(10 - 0.10) = $60.00 - $0.60 = $59.40
Setup: $(6 \times 10) - (6 \times 0.10)$
SC-6: 6 [ENTER] 10 [×] 6 [ENTER] 0.10 [×] [-]

B3: 7(x + 8) = 7x + 56
Setup: 7×8
SC-6: 7 [ENTER] 8 [×]

B4: 9(2x + 7) = 18x + 63
Setup: 9×7
SC-6: 9 [ENTER] 2 [×] 9 [ENTER] 7 [×]

B5: 4(t - 3) = 4t - 12 ($48 at t = $15)
Setup: $(4 \times 15) - (4 \times 3)$
SC-6: 4 [ENTER] 15 [×] 4 [ENTER] 3 [×] [-]

B6: 14x + 15
Setup: 3×5
SC-6: 3 [ENTER] 4 [×] 2 [+] 3 [ENTER] 5 [×]

B7: 13x + 22
Setup: $(4 \times 3) + (5 \times 2)$
SC-6: 4 [ENTER] 2 [×] 5 [+] 4 [ENTER] 3 [×] 5 [ENTER] 2 [×] [+]

B8: 3x + 5
Setup: $10 \div 2$
SC-6: 6 [ENTER] 2 [÷] 10 [ENTER] 2 [÷]

B9: 2(2x + 5) + 2(x + 3) = 6x + 16
Setup: $(2 \times 5) + (2 \times 3)$
SC-6: 2 [ENTER] 2 [×] 2 [+] 2 [ENTER] 5 [×] 2 [ENTER] 3 [×] [+]

B10: 12(15) = 180 and 120 + 60 = 180
Setup: 12×15
SC-6: 12 [ENTER] 15 [×]

SEC 4.4 (pp.43-44): FACTORING EXPRESSIONS USING THE GCF

Methods: W1: Factor Out the GCF Using Prime Factors | W2: Equal-Rows Area Model / Division Ladder

PRACTICE A (p.43): KEY IDEAS (#1-#10)

ExA: 6(3x + 4)

A1: 4(3 + 5)
Setup: GCF(12, 20)
SC-6: 12 [ENTER] 20 [GCF] -> 4

A2: 4(2x + 3)
Setup: GCF(8, 12)
SC-6: 8 [ENTER] 12 [GCF] -> 4

A3: 5(3y - 5)
Setup: GCF(15, 25)
SC-6: 15 [ENTER] 25 [GCF] -> 5

A4: 12(2x + 3)
Setup: GCF(24, 36)
SC-6: 24 [ENTER] 36 [GCF] -> 12

A5: 16(2n - 3)
Setup: GCF(32, 48)
SC-6: 32 [ENTER] 48 [GCF] -> 16

A6: 14(3a + 1)
Setup: GCF(42, 14)
SC-6: 42 [ENTER] 14 [GCF] -> 14

A7: 12(3m - 5)
Setup: GCF(36, 60)
SC-6: 36 [ENTER] 60 [GCF] -> 12

A8: 12(2x + 3) (4 and 6 still share a factor of 2)
Setup: GCF(4, 6)
SC-6: 4 [ENTER] 6 [GCF] -> 2 left!

A9: Height = 14 in.; Width = (2x + 3) in.
Setup: GCF(28, 42)
SC-6: 28 [ENTER] 42 [GCF] -> 14

A10: (9x + 12) feet
Setup: $36 \div 4$; $48 \div 4$
SC-6: 36 [ENTER] 4 [÷]; 48 [ENTER] 4 [÷]

PRACTICE B (p.44): REAL LIFE (#1-#10)

ExB: 15(3x - 2)

B1: 8(2x + 3) -> 8 bags
Setup: GCF(16, 24)
SC-6: 16 [ENTER] 24 [GCF]

B2: 7(3x + 5) -> length = 3x + 5 m
Setup: GCF(21, 35)
SC-6: 21 [ENTER] 35 [GCF]

B3: 4(5x + 9) -> side = 5x + 9 in
Setup: GCF(20, 36)
SC-6: 20 [ENTER] 36 [GCF]

B4: 3(6y - 9) -> side = 6y - 9 cm
Setup: GCF(18, 27)
SC-6: 18 [ENTER] 27 [GCF]

B5: 8(5x + 8)
Setup: GCF(40, 64)
SC-6: 40 [ENTER] 64 [GCF]

B6: 18(2s + 3) -> 18 player kits
Setup: GCF(36, 54)
SC-6: 36 [ENTER] 54 [GCF]

B7: 6(2x + 3y + 5)
Setup: GCF(GCF(12, 18), 30)
SC-6: 12 [ENTER] 18 [GCF] 30 [GCF]

B8: 7(5d - 2) -> 5d - 2 per row
Setup: GCF(35, 14)
SC-6: 35 [ENTER] 14 [GCF]

B9: 24x + 36 = 12(2x + 3)
Setup: GCF((10 + 14), (18 + 18))
SC-6: 10 [ENTER] 14 [+] 18 [ENTER] 18 [+] [GCF]

B10: 9(3x - 2); at x=2 both equal 36
Setup: GCF(27, 18)
SC-6: 27 [ENTER] 18 [GCF]

Unit 5: Areas of Polygons & Elapsed Time — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Planetary Geared Clock (Geared Modulo-60 Clock)

Target Misconception: Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.

Hands-On Block Remediation: Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock + [CLK] key for base-60 time.



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 5.1 (pp.46-47): AREAS OF PARALLELOGRAMS AND RECTANGLES

Methods: W1: Area Formula A = bh | W2: Cut-and-Slide Rectangle Decomposition

PRACTICE A (p.46): KEY IDEAS (#1-#10)

ExA: 96 cm²

A1: 135 m²
Setup: A = 15 × 9
SC-6: 15 [ENTER] 9 [f][RECT]

A2: 36 m²
Setup: A = 8 × 4.5
SC-6: 8 [ENTER] 4.5 [f][RECT]

A3: 54.4 m²
Setup: A = 8.5 × 6.4
SC-6: 8.5 [ENTER] 6.4 [f][RECT]

A4: 169 cm²
Setup: 13²
SC-6: 13 [x²]

A5: 26 in.²
Setup: A = 6 1/2 × 4
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [f][RECT]

A6: 9 cm
Setup: 108 ÷ 12
SC-6: 108 [ENTER] 12 [÷]

A7: 18 ft
Setup: 135 ÷ 7.5
SC-6: 135 [ENTER] 7.5 [÷]

A8: 60 cm² (He multiplied by the slanted side 7.5)
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

A9: 666 ft²
Setup: (9×18.5) × 4
SC-6: 9 [ENTER] 18.5 [f][RECT] 4 [×]

A10: 168 ft²
Setup: (15×12) - (4×3)
SC-6: 15 [ENTER] 12 [f][RECT] 4 [ENTER] 3 [f][RECT] [-]

PRACTICE B (p.47): REAL LIFE (#1-#10)

ExB: 12 inches

B1: 180 ft²
Setup: A = 10 × 18
SC-6: 10 [ENTER] 18 [f][RECT]

B2: 60 ft
Setup: (10 + 20) × 2
SC-6: 10 [ENTER] 20 [+] 2 [×]

B3: 14.4 ft²
Setup: A = 4.5 × 3.2
SC-6: 4.5 [ENTER] 3.2 [f][RECT]

B4: 15 inches
Setup: 135 ÷ 9
SC-6: 135 [ENTER] 9 [÷]

B5: 216 cm²
Setup: (6×4.5) × 8
SC-6: 6 [ENTER] 4.5 [f][RECT] 8 [×]

B6: 228 ft²
Setup: A = 19 × 12
SC-6: 19 [ENTER] 12 [f][RECT]

B7: 2,520 ft²
Setup: (65×40) - (10×8)
SC-6: 65 [ENTER] 40 [f][RECT] 10 [ENTER] 8 [f][RECT] [-]

B8: 22 in²
Setup: 5 1/2 × 4
SC-6: 5 [.] 1 [.] 2 [ENTER] 4 [×]

B9: Both equal 144 cm² (same base and perpendicular height)
Setup: A = 16 × 9
SC-6: 16 [ENTER] 9 [f][RECT]

B10: 28 ft² × $4 = $112.00
Setup: (8×3.5) × 4
SC-6: 8 [ENTER] 3.5 [f][RECT] 4 [×]

SEC 5.2 (pp.48-49): AREAS OF TRIANGLES

Methods: W1: Formula A = (b × h) ÷ 2 | W2: Take Half of Base or Height First (½b × h)

PRACTICE A (p.48): KEY IDEAS (#1-#10)

ExA: 30 cm²

A1: 750 cm²
Setup: A = ½(25 × 60)
SC-6: 25 [ENTER] 60 [g][TRI]

A2: 18 ft²
Setup: A = ½(9 × 4)
SC-6: 9 [ENTER] 4 [g][TRI]

A3: 88 cm²
Setup: A = ½(16 × 11)
SC-6: 16 [ENTER] 11 [g][TRI]

A4: 52.5 m²
Setup: A = ½(12.5 × 8.4)
SC-6: 12.5 [ENTER] 8.4 [g][TRI]

A5: 13 1/2 in.² (13.5)
Setup: A = ½(4 1/2 × 6)
SC-6: 4 [.] 1 [.] 2 [ENTER] 6 [g][TRI]

A6: 9 cm
Setup: (54 × 2) ÷ 12
SC-6: 54 [ENTER] 2 [×] 12 [÷]

A7: 12 m
Setup: (84 × 2) ÷ 14
SC-6: 84 [ENTER] 2 [×] 14 [÷]

A8: 20 cm² (He forgot to divide by 2)
Setup: A = ½(8 × 5)
SC-6: 8 [ENTER] 5 [g][TRI]

A9: 210 ft²
Setup: ½(12×25) + ½(8×15)
SC-6: 12 [ENTER] 25 [g][TRI] 8 [ENTER] 15 [g][TRI] [+]

A10: 1,400 cm²
Setup: A = ½(40 × 70)
SC-6: 40 [ENTER] 70 [g][TRI]

PRACTICE B (p.49): REAL LIFE (#1-#10)

ExB: 8 inches

B1: 150 ft²
Setup: A = ½(12 × 25)
SC-6: 12 [ENTER] 25 [g][TRI]

B2: 14 ft² (1 side); 28 ft² (both)
Setup: A = ½(8 × 3.5)
SC-6: 8 [ENTER] 3.5 [g][TRI]

B3: 141 ft²
Setup: A = ½(30 × 9.4)
SC-6: 30 [ENTER] 9.4 [g][TRI]

B4: 36 inches
Setup: (450 × 2) ÷ 25
SC-6: 450 [ENTER] 2 [×] 25 [÷]

B5: 6 feet
Setup: (18 × 2) ÷ 6
SC-6: 18 [ENTER] 2 [×] 6 [÷]

B6: 10.125 in² (10 1/8 in²)
Setup: A = ½(4.5 × 4.5)
SC-6: 4.5 [ENTER] 4.5 [g][TRI]

B7: 54 ft²
Setup: A = ½(9 × 12)
SC-6: 9 [ENTER] 12 [g][TRI]

B8: 630 cm² (315 cm² per wing)
Setup: ½(18×35) × 2
SC-6: 18 [ENTER] 35 [g][TRI] 2 [×]

B9: 13 in²
Setup: (6 1/2 × 4) ÷ 2
SC-6: 6 [.] 1 [.] 2 [ENTER] 4 [×] 2 [÷]

B10: 64 ft²
Setup: ½(5×6.4) × 4
SC-6: 5 [ENTER] 6.4 [g][TRI] 4 [×]

SEC 5.3 (pp.50-51): AREAS OF TRAPEZOIDS AND COMPOSITE FIGURES

Methods: W1: Decompose & Add Sub-Areas | W2: Trapezoid Formula A = ½h(b1 + b2) | W3: Enclosing Box Minus Cutouts

PRACTICE A (p.50): KEY IDEAS (#1-#10)

ExA: 126 ft²

A1: 72 cm²
Setup: A = ½((10 + 14) × 6)
SC-6: 10 [ENTER] 14 [+] 6 [g][TRI]

A2: 135 ft²
Setup: A = ½((18 + 12) × 9)
SC-6: 18 [ENTER] 12 [+] 9 [g][TRI]

A3: 92 ft²
Setup: Stats(10, 6, 8, 4)
SC-6: 10 [ENTER] 6 [f][RECT] [Σ+] 8 [ENTER] 4 [f][RECT] [Σ+] [f][Σx]

A4: 120 in.²
Setup: Stats(15, 6, 10, 6)
SC-6: 15 [ENTER] 6 [f][RECT] [Σ+] 10 [ENTER] 6 [g][TRI] [Σ+] [f][Σx]

A5: 29.4 m²
Setup: A = ½((8.5 + 5.5) × 4.2)
SC-6: 8.5 [ENTER] 5.5 [+] 4.2 [g][TRI]

A6: 56 in.²
Setup: Stats(10, 8, 6, 4)
SC-6: 10 [ENTER] 8 [f][RECT] [Σ+] 6 [ENTER] 4 [f][RECT] [f][Σ-] [f][Σx]

A7: 10 cm
Setup: (90 × 2) ÷ (11 + 7)
SC-6: 90 [ENTER] 2 [×] 11 [ENTER] 7 [+] [÷]

A8: 88 ft²
Setup: Stats(60, 5, 9, 4)
SC-6: 60 [Σ+] 5 [ENTER] 9 [+] 4 [g][TRI] [Σ+] [f][Σx]

A9: 104 cm² (80 + 24 = 104)
Setup: A = ½((10 + 16) × 8)
SC-6: 10 [ENTER] 16 [+] 8 [g][TRI]

A10: 251 ft²
Setup: Stats(300, 24, 25)
SC-6: 300 [Σ+] 24 [f][Σ-] 25 [f][Σ-] [f][Σx]

PRACTICE B (p.51): REAL LIFE (#1-#10)

ExB: 1,008 square inches

B1: 440 in²
Setup: A = ½((16 + 28) × 20)
SC-6: 16 [ENTER] 28 [+] 20 [g][TRI]

B2: 40 in² per side; 160 in² total
Setup: A = ½((4 + 6) × 8)
SC-6: 4 [ENTER] 6 [+] 8 [g][TRI]

B3: 84 + 50 = 134 ft²
Setup: (14×6) + (10×5)
SC-6: 14 [ENTER] 6 [f][RECT] 10 [ENTER] 5 [f][RECT] [+]

B4: 42 cm² (1 fin); 126 cm² (3 fins)
Setup: A = ½((5 + 9) × 6)
SC-6: 5 [ENTER] 9 [+] 6 [g][TRI]

B5: 90 + 48 = 138 in²
Setup: (15×6) + ½(12×8)
SC-6: 15 [ENTER] 6 [f][RECT] 12 [ENTER] 8 [g][TRI] [+]

B6: 1,152 - 36 = 1,116 in²
Setup: (48×24) - 6²
SC-6: 48 [ENTER] 24 [f][RECT] 6 [x²] [-]

B7: h = 8 ft
Setup: 104 ÷ 13
SC-6: 104 [ENTER] 13 [÷]

B8: 280 + 60 = 340 ft²
Setup: (20×14) + ½(20×6)
SC-6: 20 [ENTER] 14 [f][RECT] 20 [ENTER] 6 [g][TRI] [+]

B9: 416 - 200 = 216 m²
Setup: (26×16) - (20×10)
SC-6: 26 [ENTER] 16 [f][RECT] 20 [ENTER] 10 [f][RECT] [-]

B10: 42 cm²
Setup: ½((4 + 8)×3.5) × 2
SC-6: 4 [ENTER] 8 [+] 3.5 [g][TRI] 2 [×]

SEC 5.4 (pp.52-53): POLYGONS IN THE COORDINATE PLANE & ELAPSED TIME

Methods: W1: Subtract Coordinates / Regroup 60 min = 1 hr | W2: Count Grid Squares / Time-Line Jumps

PRACTICE A (p.52): KEY IDEAS (#1-#10)

ExA: Width = 7, Height = 5, Perimeter = 24, Area = 35

A1: 8 units
Setup: 11 - 3
SC-6: 11 [ENTER] 3 [-]

A2: 54 square units
Setup: A = (10 - 1) × (8 - 2)
SC-6: 10 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

A3: 10:58 AM
Setup: 7.10 + 3.48
SC-6: [CLK] 7.10 [ENTER] 3.48 [+]

A4: 2.5 hours; 4.75 hours
Setup: 2.30 (to HR); 4.45 (to HR)
SC-6: 2.30 [f][->HR]; 4.45 [f][->HR]

A5: 6 hours 50 minutes
Setup: 15.05 - 8.15
SC-6: [CLK] 15.05 [ENTER] 8.15 [-]

A6: 1 hour 45 minutes
Setup: 1.75 (to HMS)
SC-6: 1.75 [g][->HMS]

A7: 2 hours 35 minutes
Setup: 11.25 - 8.50
SC-6: [CLK] 11.25 [ENTER] 8.50 [-]

A8: 24 square units
Setup: A = ½(8 × 6)
SC-6: 8 [ENTER] 6 [g][TRI]

A9: 5.5 hours (5 hr 30 min); $101.75
Setup: 5.5 × 18.50
SC-6: 5.5 [ENTER] 18.50 [×]

A10: Perimeter = 420 yd; Area = 10,800 yd²
Setup: A = 12 × 9
SC-6: 12 [ENTER] 9 [f][RECT]

PRACTICE B (p.53): REAL LIFE (#1-#10)

ExB: 12:30 PM

B1: W=10, H=6, P=32, Area=60
Setup: A = (11 - 1) × (8 - 2)
SC-6: 11 [ENTER] 1 [-] 8 [ENTER] 2 [-] [f][RECT]

B2: Base=8, Height=6, Area=24
Setup: A = ½((10 - 2) × (7 - 1))
SC-6: 10 [ENTER] 2 [-] 7 [ENTER] 1 [-] [g][TRI]

B3: 12:15 PM
Setup: 8.35 + 3.40
SC-6: [CLK] 8.35 [ENTER] 3.40 [+]

B4: 3 hours 35 minutes
Setup: 14.20 - 10.45
SC-6: [CLK] 14.20 [ENTER] 10.45 [-]

B5: 3 hours 50 minutes
Setup: 1.52 + 1.58
SC-6: [CLK] 1.52 [ENTER] 1.58 [+]

B6: 3.5 hr × $14 = $49.00
Setup: 3.5 × 14
SC-6: 3.5 [ENTER] 14 [×]

B7: 2 hours 45 minutes
Setup: 2.75 (to HMS)
SC-6: 2.75 [g][->HMS]

B8: 4 hours 30 minutes
Setup: (1.25 + 2.15) + 0.50
SC-6: [CLK] 1.25 [ENTER] 2.15 [+] 0.50 [+]

B9: (3, 10); Area = 36
Setup: A = (9 - 3) × (10 - 4)
SC-6: 9 [ENTER] 3 [-] 10 [ENTER] 4 [-] [f][RECT]

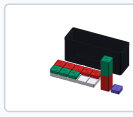
B10: 180 miles
Setup: 2.25 × 80
SC-6: 2.25 [ENTER] 80 [×]

Unit 6: Integers, Rational Numbers & The Coordinate Plane — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 6 • Hands-On Models: Ten-Frame Two-Tone Counters (Ten-Frames & Two-Tone Counters) + Algebra Zero-Pair Tiles (Algebra & Area Tiles)

Target Misconception: Students confuse the subtraction operator [-] with the negative sign [+/-], think -18 > -5 because 18 > 5, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.

Hands-On Block Remediation: Use two-tone (+1 / -1) Zero-Pair counters and Zero-Pair tiles, plot signed jumps on the SC-6 Number Line ([f] [LINE]), test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 6.1 (pp.55-56): POSITIVE AND NEGATIVE INTEGERS & OPPOSITES

Methods: W1: Equal Distance on Opposite Sides of 0 (Number Line) | W2: Additive Inverse & Zero-Pair Counters

PRACTICE A (p.55): KEY IDEAS (#1-#10)

ExA: Opposite of -15 is +15; -15 + 15 = 0

A1: -25; Opposite = +25
Setup: -25; +/-
SC-6: 25 [+/-] -> -25; [+/-] -> 25

A2: +42; and +42
Setup: 42
SC-6: 42 [+/-] [+/-]

A3: 0 m (Sea level)
Setup: -18 + 18
SC-6: 18 [+/-] [ENTER] 18 [+]

A4: 0
Setup: 6 + -6
SC-6: 6 [ENTER] 6 [+/-] [+]

A5: 0 ft
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

A6: 0 (0 is 0 units from itself)
Setup: -0
SC-6: 0 [+/-]

A7: +7°F
Setup: -12 + 19
SC-6: 12 [+/-] [ENTER] 19 [+]

A8: 0 (Even par)
Setup: -3 + 3
SC-6: 3 [+/-] [ENTER] 3 [+]

A9: Floor 3
Setup: (14 + -20) + 9
SC-6: 14 [ENTER] 20 [+/-] [+] 9 [+]

A10: +$18,800
Setup: ((14500 - 3200) + 8900) - 1400
SC-6: 14500 [ENTER] 3200 [-] 8900 [+] 1400 [-]

PRACTICE B (p.56): REAL LIFE (#1-#10)

ExB: +$5

B1: 0 ft (opposites / additive inverses)
Setup: -120 + 120
SC-6: 120 [+/-] [ENTER] 120 [+]

B2: -282 ft; opposite is +282 ft
Setup: -282
SC-6: 282 [+/-]

B3: +19 yards
Setup: (12 + -8) + 15
SC-6: 12 [ENTER] 8 [+/-] [+] 15 [+]

B4: -28°C; opposite is +28°C
Setup: -28
SC-6: 28 [+/-]

B5: +$7
Setup: -18 + 25
SC-6: 18 [+/-] [ENTER] 25 [+]

B6: -5 (5 under par)
Setup: (-4 + 2) + -3
SC-6: 4 [+/-] [ENTER] 2 [+] 3 [+/-] [+]

B7: -x = +35; -(-x) = -35
Setup: 35
SC-6: 35 [+/-] [+/-]

B8: +$10
Setup: (25 - 25) + 10
SC-6: 25 [ENTER] 25 [-] 10 [+]

B9: +35 meters
Setup: (-40 + 95) - 20
SC-6: 40 [+/-] [ENTER] 95 [+] 20 [-]

B10: +23
Setup: -14 + -9
SC-6: 14 [+/-] [ENTER] 9 [+/-] [+] [+/-]

SEC 6.2 (pp.57-58): COMPARING AND ORDERING INTEGERS & RATIONAL NUMBERS

Methods: W1: Convert to Decimals & Plot on Number Line | W2: Rewrite with Least Common Denominator (LCD)

PRACTICE A (p.57): KEY IDEAS (#1-#10)

ExA: -5 > -18

A1: -5°C (-12 < -5)
Setup: Compare -5 < -12
SC-6: 5 [+/-] [ENTER] 12 [+/-] [g][<]

A2: -9 < -2
Setup: Compare -2 < -9
SC-6: 2 [+/-] [ENTER] 9 [+/-] [g][<]

A3: -15 < 0
Setup: Compare 0 < -15
SC-6: 0 [ENTER] 15 [+/-] [g][<]

A4: -10, -7, -2, 0, 4
Setup: fLINE
SC-6: Left-to-right on [f][LINE]

A5: -3/4 < -0.6 (-0.6 is greater)
Setup: Compare -0.6 < -0.75
SC-6: 0.6 [+/-] [ENTER] 0.75 [+/-] [g][<]

A6: -2 1/2 < -2.3
Setup: Compare -2.3 < -2.5
SC-6: 2.3 [+/-] [ENTER] 2.5 [+/-] [g][<]

A7: -5/8 > -7/8
Setup: Compare -7/8 > -5/8
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [f][>]

A8: -4.1 is closer to 0 (further to the right) than -4.5
Setup: Compare -4.5 < -4.1
SC-6: 4.5 [+/-] [ENTER] 4.1 [+/-] [g][<]

A9: -1 3/4, -1.5, -0.5, 1 1/4
Setup: -1.75, -1.5, -0.5, 1.25
SC-6: Mental / Table Check

A10: Deepest: -15.2 m; Shallowest: -14 1/2 m
Setup: -15.2 < -14.8 < -14.5
SC-6: Mental / Table Check

PRACTICE B (p.58): REAL LIFE (#1-#10)

ExB: -0.75 > -4/5 (Diver B is closer to the surface)

B1: -11°C > -18°C (Tuesday was warmer)
Setup: Compare -18 > -11
SC-6: 18 [+/-] [ENTER] 11 [+/-] [f][>]

B2: Puffin (-45.8 m) is deeper; -45.25 > -45.8
Setup: Compare -45.8 > -45.25
SC-6: 45.8 [+/-] [ENTER] 45.25 [+/-] [f][>]

B3: -2/3 > -5/6 (-4/6 > -5/6)
Setup: Compare -5/6 > -2/3
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 2 [.] 3 [+/-] [f][>]

B4: -9, -6, -1, +2
Setup: Compare -6 < -9
SC-6: 6 [+/-] [ENTER] 9 [+/-] [g][<]

B5: -89°C, -39°C, -6°C, 0°C
Setup: Compare -39 < -89
SC-6: 39 [+/-] [ENTER] 89 [+/-] [g][<]

B6: Sea turtle (-3.1 > -3.25)
Setup: Compare -3.1 < -3.25
SC-6: 3.1 [+/-] [ENTER] 3.25 [+/-] [g][<]

B7: Stock B (-1.08% > -1.8%)
Setup: Compare -1.8 > -1.08
SC-6: 1.8 [+/-] [ENTER] 1.08 [+/-] [f][>]

B8: Examples: -3 1/2 and -3.5
Setup: -3.5
SC-6: 3.5 [+/-]

B9: -2, -7/4 (-1.75), -1.6, 3/2 (1.5)
Setup: -7/4 -> Dec/Frac
SC-6: [.] 7 [.] 4 [+/-] [FDISP]

B10: a is farther from 0 (e.g., -10 < -2)
Setup: |-10|
SC-6: 10 [+/-] [g][|x|]

SEC 6.3 (pp.59-60): ABSOLUTE VALUE AND MAGNITUDE

Methods: W1: Measure Distance from 0 on a Number Line | W2: Evaluate |x| Grouping Bars Before Outside Signs

PRACTICE A (p.59): KEY IDEAS (#1-#10)

ExA: |-14| = 14, |14| = 14, -|-14| = -14

A1: 125
Setup: |-125|
SC-6: 125 [+/-] [g][|x|]

A2: 30
Setup: |-18| + 12
SC-6: 18 [+/-] [g][|x|] [ENTER] 12 [+]

A3: 8.0
Setup: |-4.8| + |-3.2|
SC-6: 4.8 [+/-] [g][|x|] [ENTER] 3.2 [+/-] [g][|x|] [+]

A4: 16
Setup: |-35| - |-19|
SC-6: 35 [+/-] [g][|x|] [ENTER] 19 [+/-] [g][|x|] [-]

A5: |-12| > |9| (12 > 9)
Setup: 12 > 9
SC-6: Mental / Table Check

A6: -24
Setup: |-24|
SC-6: 24 [+/-] [g][|x|] [+/-]

A7: 32
Setup: 50 - |-18|
SC-6: 50 [ENTER] 18 [+/-] [g][|x|] [-]

A8: -|-9|, -6, 0, |-3|, |-8|
Setup: Values: 8, -6, 3, 0, -9
SC-6: Mental / Table Check

A9: Seagull is closer by 7 feet (25 < 32)
Setup: |-32| - 25
SC-6: 32 [+/-] [g][|x|] [ENTER] 25 [-]

A10: Greater debt: -$45 ($45); Greater balance: -$28
Setup: |-45|=45 > |-28|=28
SC-6: Mental / Table Check

PRACTICE B (p.60): REAL LIFE (#1-#10)

ExB: 46 m plunge; and 50 - |-18| = 32

B1: |32| + |-14| = 46 meters
Setup: |32| + |-14|
SC-6: 32 [g][|x|] 14 [+/-] [g][|x|] [+]

B2: -$40 is greater balance; Account A owes more ($65)
Setup: |-65|
SC-6: 65 [+/-] [g][|x|]

B3: |-18| + |14| = 32 meters
Setup: |-18| + |14|
SC-6: 18 [+/-] [g][|x|] 14 [g][|x|] [+]

B4: 35 and -35 (outside minus applies after | |)
Setup: |-35|
SC-6: 35 [+/-] [g][|x|] [+/-]

B5: -1.5 cm (|-1.5| = 1.5 cm is smallest)
Setup: |-1.5|
SC-6: 1.5 [+/-] [g][|x|]

B6: 42 ÷ 6 + 11 = 18
Setup: (|-42| ÷ |-6|) + |-11|
SC-6: 42 [+/-] [g][|x|] 6 [+/-] [g][|x|] [÷] 11 [+/-] [g][|x|] [+]

B7: |25| + |-140| = 165 feet
Setup: |25| + |-140|
SC-6: 25 [g][|x|] 140 [+/-] [g][|x|] [+]

B8: 19 and -19
Setup: |-19|
SC-6: 19 [+/-] [g][|x|]

B9: -|-9| (-9), -6, |5| (5), |-8| (8)
Setup: |-9|
SC-6: 9 [+/-] [g][|x|] [+/-]

B10: 4 5/8
Setup: (3 1/2 + 2 1/4) - 1 1/8
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+] 1 [.] 1 [.] 8 [-]

SEC 6.4 (pp.61-62): THE FOUR-QUADRANT COORDINATE PLANE & REFLECTIONS

Methods: W1: Plot & Count Distance to the Mirror Axis | W2: Coordinate Reflection Sign Rules

PRACTICE A (p.61): KEY IDEAS (#1-#10)

ExA: A: Quadrant II, B: Quadrant IV, C: Quadrant III

A1: Quadrant IV
Setup: (+, -) -> Right, Down
SC-6: Mental / Table Check

A2: Quadrant III
Setup: (-, -) -> Left, Down
SC-6: Mental / Table Check

A3: Quadrant II
Setup: (-, +) -> Left, Up
SC-6: Mental / Table Check

A4: (-6, 9)
Setup: -6
SC-6: 6 [+/-] -> -6

A5: (6, -9)
Setup: -9
SC-6: 9 [+/-] -> -9

A6: (-3, 7)
Setup: 7
SC-6: 7 [+/-] [+/-] -> +7

A7: (8, -4)
Setup: Flip x -> (8,4); flip y -> (8,-4)
SC-6: Mental / Table Check

A8: (0, -9) on y-axis; (-7, 0) on x-axis
Setup: 0 sideways vs 0 up/down
SC-6: Mental / Table Check

A9: (-3, 3) in Quadrant II
Setup: 4 - 7; -5 + 8
SC-6: 4 [ENTER] 7 [-]; 5 [+/-] [ENTER] 8 [+]

A10: D(-4, -5) in Quadrant III
Setup: Match x of A (-4) & y of C (-5)
SC-6: Mental / Table Check

PRACTICE B (p.62): REAL LIFE (#1-#10)

ExB: (5, -8) in Quadrant IV

B1: Cave: II, Lighthouse: I, Shipwreck: III, Pier: IV
Setup: 4
SC-6: 8 [+/-] [ENTER] 4

B2: (0, -7) is on y-axis; (9, 0) is on x-axis
Setup: -7
SC-6: 0 [ENTER] 7 [+/-]

B3: (8, -12) in Quadrant IV
Setup: -12
SC-6: 8 [ENTER] 12 [+/-]

B4: (9, 5) in Quadrant I
Setup: 5
SC-6: 9 [+/-] [+/-] [ENTER] 5

B5: Step 1: (-4, 9); Final: (4, 9) in Quad I
Setup: 9
SC-6: 4 [+/-] [+/-] [ENTER] 9 [+/-] [+/-]

B6: (4, -8) in Quadrant IV
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

B7: P: Quadrant II; Q: Quadrant III
Setup: 4.5
SC-6: 2.5 [+/-] [ENTER] 4.5

B8: y-axis (x-coordinate changed sign, y stayed -3)
Setup: -7
SC-6: 7 [+/-]

B9: (-5, 5), (-5, -5), and (5, -5)
Setup: -5
SC-6: 5 [+/-]

B10: (x, y) is in Quad II; (-x, -y) is in Quad IV
Setup: -1
SC-6: 1 [+/-]

SEC 6.5 (pp.63-64): DISTANCES IN THE COORDINATE PLANE

Methods: W1: Add/Subtract Absolute Values to the Axis | W2: Absolute Difference of Coordinates |x1 - x2|

PRACTICE A (p.63): KEY IDEAS (#1-#10)

ExA: 15 units; and 8 units

A1: 27 units
Setup: |(-12 - 15)|
SC-6: 12 [+/-] [ENTER] 15 [-] [g][|x|]

A2: 11 units
Setup: |(-4 - -15)|
SC-6: 4 [+/-] [ENTER] 15 [+/-] [-] [g][|x|]

A3: 15 units
Setup: |(9 - -6)|
SC-6: 9 [ENTER] 6 [+/-] [-] [g][|x|]

A4: 9 units
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

A5: 10 units
Setup: |(-3.5 - 6.5)|
SC-6: 3.5 [+/-] [ENTER] 6.5 [-] [g][|x|]

A6: 8 units
Setup: |(-4.5 - 3.5)|
SC-6: 4.5 [+/-] [ENTER] 3.5 [-] [g][|x|]

A7: (4, 3)
Setup: -6 + 10
SC-6: 6 [+/-] [ENTER] 10 [+]

A8: 13 units (They lie on opposite sides of the y-axis: |-5| + |..
Setup: |(-5 - 8)|
SC-6: 5 [+/-] [ENTER] 8 [-] [g][|x|]

A9: Perimeter = 34 units, Area = 72 sq units
Setup: Width 9, Height 8
SC-6: Mental / Table Check

A10: 20 grid units = 10 miles
Setup: 10 + 10 = 20 units × 0.5
SC-6: Mental / Table Check

PRACTICE B (p.64): REAL LIFE (#1-#10)

ExB: Perimeter = 34 units; Area = 72 square units

B1: |-8| + |11| = 19 units
Setup: |(-8 - 11)|
SC-6: 8 [+/-] [ENTER] 11 [-] [g][|x|]

B2: 9 units = 90 meters
Setup: |(-14 - -5)|
SC-6: 14 [+/-] [ENTER] 5 [+/-] [-] [g][|x|]

B3: |15| + |-35| = 50 units
Setup: |(15 - -35)|
SC-6: 15 [ENTER] 35 [+/-] [-] [g][|x|]

B4: W = 10, H = 6, Perimeter = 32 units
Setup: (|(-3 - 7)| + |(4 - -2)|) × 2
SC-6: 3 [+/-] [ENTER] 7 [-] [g][|x|] 4 [ENTER] 2 [+/-] [-] [g][|x|]..

B5: 60 square units
Setup: A = 10 × 6
SC-6: 10 [ENTER] 6 [f][RECT]

B6: 7.75 miles
Setup: 3.5 + 4.25
SC-6: 3.5 [ENTER] 4.25 [+]

B7: 5 3/4 units
Setup: 3 1/2 + 2 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 2 [.] 1 [.] 4 [+]

B8: 7 + 10 = 17 units
Setup: (5 + 2) + (4 + 6)
SC-6: 5 [ENTER] 2 [+] 4 [ENTER] 6 [+] [+]

B9: (11, 4) and (-5, 4)
Setup: 3 - 8
SC-6: 3 [ENTER] 8 [+] 3 [ENTER] 8 [-]

B10: Base = 10, Height = 8, Area = 40 units²
Setup: A = ½((4 + 6) × (7 + 1))
SC-6: 4 [ENTER] 6 [+] 7 [ENTER] 1 [+] [g][TRI]

Unit 7: Equations & Inequalities — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 7 • Hands-On Models: Algebra Boxes & Stage (Algebra & Area Tiles) + Counting & Array Tiles (Counting & Array Tiles)

Target Misconception: Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of x + 14 = 39, and confuse open vs. solid inequality circles.

Hands-On Block Remediation: Model '=' as a Balance Scale using Algebra Boxes & Stage. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [!=] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [>=] and [g] [<=] (which flash YES or NO).



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 7.1 (pp.66-67): WRITING EQUATIONS IN ONE VARIABLE & TESTING SOLUTIONS

Methods: W1: Substitute Candidates to Test Equality | W2: Balance Scale / Mental Unwrapping

PRACTICE A (p.66): KEY IDEAS (#1-#10)

ExA: x = 9 is the solution (41 = 41)

A1: x = 8 (5(8)-7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]

A2: y = 15 (15÷3 + 4 = 9)
Setup: (15 ÷ 3) + 4
SC-6: 9 [ENTER] 15 [ENTER] 3 [÷] 4 [+] [!=]

A3: x + 14 = 52; x = 38
Setup: x + 14 = 52
SC-6: Mental / Table Check

A4: 6k = 84; k = 14
Setup: 6k = 84
SC-6: Mental / Table Check

A5: w - 9 = 17; w = 26
Setup: w - 9 = 17
SC-6: Mental / Table Check

A6: m / 5 = 13; m = 65
Setup: m / 5 = 13
SC-6: Mental / Table Check

A7: 4p + 6 = 54; p = $12
Setup: (4 × 12) + 6
SC-6: 54 [ENTER] 4 [ENTER] 12 [×] 6 [+] [!=]

A8: x - 12 = 20; x = 32
Setup: x - 12 = 20
SC-6: Mental / Table Check

A9: 10 + 3m = 28; m = 6 movies
Setup: (28 - 10) ÷ 3
SC-6: 28 [ENTER] 10 [-] 3 [÷]

A10: b / 3 = 14.50; b = $43.50
Setup: 14.50 × 3
SC-6: 14.50 [ENTER] 3 [×]

PRACTICE B (p.67): REAL LIFE (#1-#10)

ExB: n - 8 = 25; n = 33 drones

B1: x = 8 (5·8 - 7 = 33)
Setup: (5 × 8) - 7
SC-6: 33 [ENTER] 5 [ENTER] 8 [×] 7 [-] [!=]

B2: 15t + 5 = 65; yes, t = 4 is a solution
Setup: (15 × 4) + 5
SC-6: 65 [ENTER] 15 [ENTER] 4 [×] 5 [+] [!=]

B3: k - 12 = 29; k = 41
Setup: 29 + 12
SC-6: 29 [ENTER] 12 [+]

B4: n ÷ 6 = 14; n = 84
Setup: 14 × 6
SC-6: 14 [ENTER] 6 [×]

B5: 3s = 54; s = 18 cm
Setup: 54 ÷ 3
SC-6: 54 [ENTER] 3 [÷]

B6: m = 15 (100 - 60 = 40)
Setup: 100 - (4 × 15)
SC-6: 40 [ENTER] 100 [ENTER] 4 [ENTER] 15 [×] [-] [!=]

B7: p ÷ 8 = 3; p = 24 slices
Setup: 24 ÷ 8
SC-6: 24 [ENTER] 8 [÷]

B8: 45 + d = 120; d = $75
Setup: 120 - 45
SC-6: 120 [ENTER] 45 [-]

B9: Yes (both sides equal 28)
Setup: (5 × 6) - 2
SC-6: 3 [ENTER] 6 [×] 10 [+] 5 [ENTER] 6 [×] 2 [-] [!=]

B10: b - 4 = 13; b = 17 years old
Setup: 13 + 4
SC-6: 13 [ENTER] 4 [+]

SEC 7.2 (pp.68-69): SOLVING EQUATIONS USING ADDITION OR SUBTRACTION

Methods: W1: Properties of Equality (Inverse Operations) | W2: Balance Scale / Part-Part-Whole Bar Model

PRACTICE A (p.68): KEY IDEAS (#1-#10)

ExA: x = 45

A1: x = 45
Setup: 73 - 28
SC-6: 73 [ENTER] 28 [-]

A2: y = 92
Setup: 58 + 34
SC-6: 58 [ENTER] 34 [+]

A3: n = 45
Setup: 64 - 19
SC-6: 64 [ENTER] 19 [-]

A4: k = 76
Setup: 29 + 47
SC-6: 29 [ENTER] 47 [+]

A5: w = 54.4
Setup: 34.8 + 19.6
SC-6: 34.8 [ENTER] 19.6 [+]

A6: p = 16.65
Setup: 25.4 - 8.75
SC-6: 25.4 [ENTER] 8.75 [-]

A7: m = 11/24
Setup: 5/6 - 3/8
SC-6: [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-]

A8: a = 4 3/4
Setup: 3 1/2 + 1 1/4
SC-6: 3 [.] 1 [.] 2 [ENTER] 1 [.] 1 [.] 4 [+]

A9: m - 18.50 = 26.75; m = $45.25
Setup: 26.75 + 18.50
SC-6: 26.75 [ENTER] 18.50 [+]

A10: h + 42.5 = 110; h = 67.5 ft
Setup: 110 - 42.5
SC-6: 110 [ENTER] 42.5 [-]

PRACTICE B (p.69): REAL LIFE (#1-#10)

ExB: w = 54.4 g; m = 11/24

B1: s + 45 = 210; s = 165 points
Setup: 210 - 45
SC-6: 210 [ENTER] 45 [-]

B2: p - 18 = 67; p = $85
Setup: 67 + 18
SC-6: 67 [ENTER] 18 [+]

B3: h + 1450 = 5200; h = 3,750 feet
Setup: 5200 - 1450
SC-6: 5200 [ENTER] 1450 [-]

B4: f - 64.8 = 185.2; f = 250.0 grams
Setup: 185.2 + 64.8
SC-6: 185.2 [ENTER] 64.8 [+]

B5: t - 0.48 = 12.65; t = 13.13 seconds
Setup: 12.65 + 0.48
SC-6: 12.65 [ENTER] 0.48 [+]

B6: m + 3/8 = 7/8; m = 1/2 cup
Setup: 7/8 - 3/8
SC-6: [.] 7 [.] 8 [ENTER] [.] 3 [.] 8 [-]

B7: 2 3/4 + d = 4 1/2; d = 1 3/4 miles
Setup: 4 1/2 - 2 3/4
SC-6: 4 [.] 1 [.] 2 [ENTER] 2 [.] 3 [.] 4 [-]

B8: b + 38 = 125; b = 87°
Setup: 125 - 38
SC-6: 125 [ENTER] 38 [-]

B9: -40 + d = -115; d = -75 meters
Setup: -115 - -40
SC-6: 115 [+/-] [ENTER] 40 [+/-] [-]

B10: L = 44 cm; 1 length = 22 cm
Setup: (64 - 20) ÷ 2
SC-6: 64 [ENTER] 20 [-] 2 [÷]

SEC 7.3 (pp.70-71): SOLVING EQUATIONS USING MULTIPLICATION OR DIVISION

Methods: W1: Divide/Multiply by Coefficient | W2: Multiply by the Reciprocal | W3: Equal-Groups Tape Diagram

PRACTICE A (p.70): KEY IDEAS (#1-#10)

ExA: x = 17; k = 101.5

A1: x = 17
Setup: 136 ÷ 8
SC-6: 136 [ENTER] 8 [÷]

A2: y = 15
Setup: 180 ÷ 12
SC-6: 180 [ENTER] 12 [÷]

A3: m = 114
Setup: 19 × 6
SC-6: 19 [ENTER] 6 [×]

A4: k = 101.5
Setup: 14.5 × 7
SC-6: 14.5 [ENTER] 7 [×]

A5: w = 14
Setup: 63 ÷ 4.5
SC-6: 63 [ENTER] 4.5 [÷]

A6: x = 24
Setup: 18 ÷ 3/4
SC-6: 18 [ENTER] [.] 3 [.] 4 [÷]

A7: n = 40
Setup: 16 ÷ 2/5
SC-6: 16 [ENTER] [.] 2 [.] 5 [÷]

A8: p = 4/5 (0.8)
Setup: 2/3 ÷ 5/6
SC-6: [.] 2 [.] 3 [ENTER] [.] 5 [.] 6 [÷]

A9: 1.5t = 27; t = 18 minutes
Setup: 27 ÷ 1.5
SC-6: 27 [ENTER] 1.5 [÷]

A10: 3/5 s = 27; s = 45 students
Setup: 27 ÷ 3/5
SC-6: 27 [ENTER] [.] 3 [.] 5 [÷]

PRACTICE B (p.71): REAL LIFE (#1-#10)

ExB: 40 songs

B1: 12s = 168; s = $14 per shirt
Setup: 168 ÷ 12
SC-6: 168 [ENTER] 12 [÷]

B2: p ÷ 9 = 3; p = 27 slices
Setup: 3 × 9
SC-6: 3 [ENTER] 9 [×]

B3: 65t = 325; t = 5 hours
Setup: 325 ÷ 65
SC-6: 325 [ENTER] 65 [÷]

B4: 2.5p = 16.25; p = $6.50 per pound
Setup: 16.25 ÷ 2.5
SC-6: 16.25 [ENTER] 2.5 [÷]

B5: s ÷ 4 = 9; s = $36
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

B6: (3/4)c = 21; c = 28 students
Setup: 21 ÷ 3/4
SC-6: 21 [ENTER] [.] 3 [.] 4 [÷]

B7: (5/6)p = 150; p = $180
Setup: 150 ÷ 5/6
SC-6: 150 [ENTER] [.] 5 [.] 6 [÷]

B8: 13.5w = 108; w = 8 feet
Setup: 108 ÷ 13.5
SC-6: 108 [ENTER] 13.5 [÷]

B9: h ÷ 15 = 3.2; h = 48 feet
Setup: 3.2 × 15
SC-6: 3.2 [ENTER] 15 [×]

B10: 0.40n = 56; n = 140
Setup: 56 ÷ 0.40
SC-6: 56 [ENTER] 0.40 [÷]

SEC 7.4 (pp.72-73): WRITING AND GRAPHING INEQUALITIES

Methods: W1: Inverse Property of Inequality & Number Line Graph | W2: Boundary & Test-Point Substitution

PRACTICE A (p.72): KEY IDEAS (#1-#10)

ExA: x >= 19 (Solid dot at 19, arrow right)

A1: x > 13 (Open circle at 13, arrow right)
Setup: 21 - 8
SC-6: 21 [ENTER] 8 [-]

A2: y <= 24 (Solid dot at 24, arrow left)
Setup: 15 + 9
SC-6: 15 [ENTER] 9 [+]

A3: n >= 13 (Solid dot at 13, arrow right)
Setup: 78 ÷ 6
SC-6: 78 [ENTER] 6 [÷]

A4: m < 36 (Open circle at 36, arrow left)
Setup: 9 × 4
SC-6: 9 [ENTER] 4 [×]

A5: h >= 48
Setup: 'At least' = >=
SC-6: Mental / Table Check

A6: w <= 2,000
Setup: 'At most' = <=
SC-6: Mental / Table Check

A7: Only 6 (At n = 5, 23 > 23 is false)
Setup: Compare 23 > ((4 × 6) + 3)
SC-6: 23 [ENTER] 4 [ENTER] 6 [×] 3 [+] [f][>]

A8: 5 and 6
Setup: (4 × 5) - 3
SC-6: 17 [ENTER] 4 [ENTER] 5 [×] 3 [-] [f][>=]

A9: 8b <= 50; at most 6 books ($2 left over)
Setup: 50 ÷ 8 (Q & R)
SC-6: 50 [ENTER] 8 [g][÷R]

A10: p + 340 >= 500; p >= 160 points
Setup: 500 - 340
SC-6: 500 [ENTER] 340 [-]

PRACTICE B (p.73): REAL LIFE (#1-#10)

ExB: g <= 15 (at most 15 gold bars)

B1: h >= 54 (solid dot at 54, shade right); at least 5 in
Setup: 54 - 49
SC-6: 54 [ENTER] 49 [-]

B2: p + 5 <= 12 -> p <= 7 passengers
Setup: 12 - 5
SC-6: 12 [ENTER] 5 [-]

B3: 22.50 + x >= 35 -> x >= $12.50
Setup: 35 - 22.50
SC-6: 35 [ENTER] 22.50 [-]

B4: 3.50g <= 28 -> g <= 8 games
Setup: 28 ÷ 3.50
SC-6: 28 [ENTER] 3.50 [÷]

B5: 15r > 240 -> r > 16 (at least 17 rows)
Setup: 240 ÷ 15
SC-6: 240 [ENTER] 15 [÷]

B6: b - 25 < 15 -> b < 40%
Setup: 15 + 25
SC-6: 15 [ENTER] 25 [+]

B7: 8 and 10
Setup: (5 × 8) - 10
SC-6: 30 [ENTER] 5 [ENTER] 8 [×] 10 [-] [f][>=]

B8: 2c - 30 >= 70 -> c >= 50 cupcakes
Setup: (70 + 30) ÷ 2
SC-6: 70 [ENTER] 30 [+] 2 [÷]

B9: s <= 20 (solid dot at 20, shade left down to 0)
Setup: Compare 20 <= 20
SC-6: 20 [ENTER] 20 [g][<=]

B10: 45c + 170 <= 800 -> c <= 14 crates
Setup: (800 - 170) ÷ 45
SC-6: 800 [ENTER] 170 [-] 45 [÷]

SEC 7.5 (pp.74-75): WRITING EQUATIONS IN TWO VARIABLES (Y = KX AND Y = X + B)

Methods: W1: Test Constant Ratio y/x vs. Constant Difference y - x | W2: Input-Output Table & Coordinate Graph

PRACTICE A (p.74): KEY IDEAS (#1-#10)

ExA: Table A: y = 7x; Table B: y = x - 4

A1: y = 8x; y = 80
Setup: 16 ÷ 2; 8×10
SC-6: 16 [ENTER] 2 [÷] -> 8; 8×10

A2: y = x + 8; y = 28
Setup: 9 - 1; 20+8
SC-6: 9 [ENTER] 1 [-] -> +8; 20+8

A3: y = 7x; y = 84
Setup: 14 ÷ 2; 7×12
SC-6: 14 [ENTER] 2 [÷] -> 7; 7×12

A4: y = x - 4; y = 26
Setup: 30 - 4
SC-6: 30 [ENTER] 4 [-]

A5: (1, 6), (2, 12), (3, 18), (4, 24)
Setup: Repeat(+6)
SC-6: 6 [g][REP] [+] [+] [+]

A6: Independent: t; Dependent: d
Setup: d depends on t
SC-6: Mental / Table Check

A7: 292.5 miles
Setup: 65 × 4.5
SC-6: 65 [ENTER] 4.5 [×]

A8: y = 4x passes through (0, 0); y = x + 4 crosses at (0, 4)
Setup: Test x = 0 in both
SC-6: Mental / Table Check

A9: 8 hours
Setup: 100 ÷ 12.50
SC-6: 100 [ENTER] 12.50 [÷]

A10: c = 15m + 25; $205
Setup: (15 × 12) + 25
SC-6: 15 [ENTER] 12 [×] 25 [+]

PRACTICE B (p.75): REAL LIFE (#1-#10)

ExB: p = 12.50h; $81.25

B1: y = x + 5 is additive; y = 8m is multiplicative
Setup: 8 × 1
SC-6: 8 [ENTER] 1 [×]

B2: g = 18h (h = indep, g = dep); 81 grams
Setup: 18 × 4.5
SC-6: 18 [ENTER] 4.5 [×]

B3: m = z + 4; m = 12, 16, 19
Setup: 8 + 4
SC-6: 8 [ENTER] 4 [+]

B4: s = 15p; p = 12 sheets
Setup: 180 ÷ 15
SC-6: 180 [ENTER] 15 [÷]

B5: y = x - 6; $18, $29, and $44
Setup: 24 - 6
SC-6: 24 [ENTER] 6 [-]

B6: y = 3.5x; 140 wheel turns
Setup: 3.5 × 40
SC-6: 3.5 [ENTER] 40 [×]

B7: y = 9x; at x = 20, y = 180
Setup: (27 ÷ 3) × 20
SC-6: 27 [ENTER] 3 [÷] 20 [×]

B8: c = 20g + 7; $127
Setup: (20 × 6) + 7
SC-6: 20 [ENTER] 6 [×] 7 [+]

B9: Multiplicative y = kx (when x = 0, y = k·0 = 0)
Setup: 0 × 7
SC-6: 0 [ENTER] 7 [×]

B10: d = 0.2s; 3 miles away
Setup: 0.2 × 15
SC-6: 0.2 [ENTER] 15 [×]

Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + 3x1 Storage Boxes

Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 8.1 (pp.77-78): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS

Methods: W1: Unfold into a 6-Face 2D Net | W2: Opposite Face Pairs SA = 2(lw + lh + wh) | W3: Lateral Strip Ph + 2B

PRACTICE A (p.77): KEY IDEAS (#1-#10)

ExA: 158 cm²

A1: 96 cm²
Setup: $4^2 \times 6$
SC-6: 4 [x²] 6 [×]

A2: 486 in.²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

A3: 108 cm²
Setup: 2(24)+2(18)+2(12)
SC-6: Mental / Table Check

A4: 160 in.²
Setup: 2(50)+2(20)+2(10)
SC-6: Mental / Table Check

A5: 392 ft²
Setup: 2(96)+2(60)+2(40)
SC-6: Mental / Table Check

A6: 85 m²
Setup: 2(20)+2(12.5)+2(10)
SC-6: Mental / Table Check

A7: One face = 25 cm²; s = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

A8: 94 cm² (He only summed 3 faces instead of doubling for all 6..
Setup: 47 × 2
SC-6: 47 [ENTER] 2 [×]

A9: 2,100 cm²
Setup: 600 + 2(450) + 2(300)
SC-6: Mental / Table Check

A10: 500 in.² needed; Yes, with 50 in.² to spare
Setup: 2(150)+2(60)+2(40)
SC-6: Mental / Table Check

PRACTICE B (p.78): REAL LIFE (#1-#10)

ExB: Cube = 294 in²; Open-Top Bin = 260 cm²

B1: 2(160 + 600 + 240) = 2,000 cm²
Setup: (((20 × 8) + (20 × 30)) + (8 × 30)) × 2
SC-6: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [ENTER] 30 [×] [+] 2..

B2: 1 face = 81 in²; Total SA = 486 in²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

B3: 288 + 720 + 360 = 1,368 in²
Setup: ((24 × 12) + ((24 × 15) × 2)) + ((12 × 15) × 2)
SC-6: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [+] 12 [ENTER] 15 [×..

B4: SA = 500 in²; yes (100 in² to spare)
Setup: (((15 × 10) + (15 × 4)) + (10 × 4)) × 2
SC-6: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [ENTER] 4 [×] [+] 2..

B5: 120 + 192 + 160 = 472 ft²
Setup: ((12 × 10) + ((12 × 8) × 2)) + ((10 × 8) × 2)
SC-6: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+] 10 [ENTER] 8 [×]..

B6: 1 face = 25 cm²; edge = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

B7: 32 + 8 = 40 ft²
Setup: ((4 × 2) × 4) + ((2 × 2) × 2)
SC-6: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×] [+]

B8: 2(24 + 60 + 40) = 248 cm²
Setup: (((6 × 4) + (6 × 10)) + (4 × 10)) × 2
SC-6: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENTER] 10 [×] [+] 2 [×]

B9: 54 cm² vs. 216 cm² (4 times as great)
Setup: (6^2 × 6) ÷ 54
SC-6: 6 [x²] 6 [×] 54 [÷]

B10: 160 cm²
Setup: $4^2 \times 10$
SC-6: 4 [x²] 10 [×]

SEC 8.2 (pp.79-80): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS

Methods: W1: Add All Individual Faces on the 2D Net | W2: Base Area + Combined Lateral Area

PRACTICE A (p.79): KEY IDEAS (#1-#10)

ExA: 340 m²

A1: 96 cm²
Setup: $6^2 + (\frac{1}{2}(6\times5) \times 4)$
SC-6: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]

A2: 224 in.²
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

A3: 360 ft²
Setup: $12^2 + (\frac{1}{2}(12\times9) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]

A4: 62.4 cm²
Setup: $\frac{1}{2}(6\times5.2) \times 4$
SC-6: 6 [ENTER] 5.2 [g][TRI] 4 [×]

A5: 132 cm²
Setup: $\frac{1}{2}(4\times3) \times 2$
SC-6: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)

A6: 150 ft²
Setup: $\frac{1}{2}(8\times3) \times 2$
SC-6: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)

A7: 110.88 cm²
Setup: $\frac{1}{2}(8\times6.93) \times 4$
SC-6: 8 [ENTER] 6.93 [g][TRI] 4 [×]

A8: Triangular prism: 2 triangles + 3 rectangles; Square pyramid..
Setup: Count net faces
SC-6: Mental / Table Check

A9: 152 ft²
Setup: 24 + 48 + 80
SC-6: Mental / Table Check

A10: 108 ft²
Setup: $\frac{1}{2}(9\times6) \times 4$
SC-6: 9 [ENTER] 6 [g][TRI] 4 [×]

PRACTICE B (p.80): REAL LIFE (#1-#10)

ExB: 288 square feet

B1: 24 + 80 + 48 = 152 ft²
Setup: $(\frac{1}{2}(6\times4) \times 2) + ((5\times8) \times 2)) + (6\times8)$
SC-6: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][RECT] 2 [×] [+] 6 [..

B2: 144 + 240 = 384 cm²
Setup: $12^2 + (\frac{1}{2}(12\times10) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]

B3: 96 ft²
Setup: $\frac{1}{2}(8\times6) \times 4$
SC-6: 8 [ENTER] 6 [g][TRI] 4 [×]

B4: 13.8 + 240 = 253.8 cm²
Setup: (6.9 × 2) + ((4×20) × 3)
SC-6: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3 [×] [+]

B5: 48 + 120 = 168 in²
Setup: $(\frac{1}{2}(8\times6) \times 2) + (((8 + 6) + 10) \times 5)$
SC-6: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+] 10 [+] 5 [×] [+]

B6: 15.6 cm²
Setup: $\frac{1}{2}(3\times2.6) \times 4$
SC-6: 3 [ENTER] 2.6 [g][TRI] 4 [×]

B7: 96 + 600 = 696 ft²
Setup: $(\frac{1}{2}(16\times6) \times 2) + ((10\times30) \times 2)$
SC-6: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [f][RECT] 2 [×] [+]

B8: 15 in² per face; slant height = 5 in
Setup: (96 - 36) ÷ 4
SC-6: 96 [ENTER] 36 [-] 4 [÷]

B9: 96 + 480 = 576 cm²
Setup: $(\frac{1}{2}(12\times8) \times 2) + (((10 + 10) + 12) \times 15)$
SC-6: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [+] 12 [+] 15 [×] [+]

B10: Pyramid A (260 cm² > 224 cm²)
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [x²] 8 [ENTER] 10 [g..

SEC 8.3 (pp.81-82): VOLUMES OF RECTANGULAR PRISMS

Methods: W1: Base Area × Height (V = Bh) | W2: Multiply Length × Width × Height (Fractions or Decimals)

PRACTICE A (p.81): KEY IDEAS (#1-#10)

ExA: Crate = 480 in.³; Aquarium = 10 1/2 ft³

A1: 3,024 cm³
Setup: Prism(18×6×28)
SC-6: 18 [ENTER] 6 [ENTER] 28 [f][PRISM]

A2: 125 in.³
Setup: 5^3
SC-6: 5 [x³]

A3: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

A4: 224 m³
Setup: 32 × 7
SC-6: 32 [ENTER] 7 [×]

A5: 10 1/2 ft³ (10.5)
Setup: Prism(3.5×1.5×2)
SC-6: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]

A6: 27 in.³
Setup: Prism(4.5×3×2)
SC-6: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]

A7: 6 in.
Setup: 270 ÷ (9 × 5)
SC-6: 270 [ENTER] 9 [ENTER] 5 [×] [÷]

A8: 8 half-inch cubes
Setup: 2^3
SC-6: 2 [x³]

A9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

A10: 216 half-inch cubes
Setup: 27 × 2^3
SC-6: 27 [ENTER] 2 [x³] [×]

PRACTICE B (p.82): REAL LIFE (#1-#10)

ExB: 8 inches deep

B1: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

B2: 48 ft³
Setup: Prism(8×4×1.5)
SC-6: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]

B3: 15,625 cm³
Setup: 25^3
SC-6: 25 [x³]

B4: 65 in³
Setup: Prism(6.5×4×2.5)
SC-6: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]

B5: 8 per in³; 96 half-inch cubes
Setup: 12 × 8
SC-6: 12 [ENTER] 8 [×]

B6: 3 cm rise
Setup: 120 ÷ 40
SC-6: 120 [ENTER] 40 [÷]

B7: 77 ft³
Setup: Prism(5.5×4×3.5)
SC-6: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]

B8: 36 ft³
Setup: (12 × 9) ÷ 3
SC-6: 12 [ENTER] 9 [×] 3 [÷]

B9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

B10: 4,608 in³ ~ 19.95 (about 20) gallons
Setup: Prism(24×12×16) ÷ 231
SC-6: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

SEC 8.4 (pp.83-84): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM

Methods: W1: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | W2: Decimal Approximation (π ~ 3.14) & Scaled Triples

PRACTICE A (p.83): KEY IDEAS (#1-#10)

ExA: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

A1: 78.54 cm²
Setup: $\pi \times 5^2$
SC-6: 5 [f][CIRC]

A2: 706.86 cm²
Setup: $\pi \times 15^2$
SC-6: 15 [f][CIRC]

A3: 5 cm
Setup: $\sqrt{(3^2 + 4^2)}$
SC-6: 3 [ENTER] 4 [g][HYP]

A4: 10 meters
Setup: $\sqrt{(6^2 + 8^2)}$
SC-6: 6 [ENTER] 8 [g][HYP]

A5: 13 ft
Setup: $\sqrt{(12^2 + 5^2)}$
SC-6: 12 [ENTER] 5 [g][HYP]

A6: r = 8 m; Area = 201.06 m²
Setup: $\pi \times (16 \div 2)^2$
SC-6: 16 [ENTER] 2 [÷] [f][CIRC]

A7: 18.85 m³
Setup: $1/3\pi(3^2)(2)$
SC-6: 3 [ENTER] 2 [g][CONE]

A8: 188.50 in.³
Setup: $1/3\pi(6^2)(5)$
SC-6: 6 [ENTER] 5 [g][CONE]

A9: Cylinder = 282.74 cm³; Cone = 94.25 cm³
Setup: $1/3\pi(3^2)(10)$
SC-6: 3 [ENTER] 10 [g][CONE]

A10: 40 inches
Setup: $\sqrt{(32^2 + 24^2)}$
SC-6: 32 [ENTER] 24 [g][HYP]

PRACTICE B (p.84): REAL LIFE (#1-#10)

ExB: 94.25 cm² (exact: 30π cm²)

B1: 153.94 ft²
Setup: $\pi \times 7^2$
SC-6: 7 [f][CIRC]

B2: r = 20 m; Area ~ 1,256.64 m²
Setup: $\pi \times 20^2$
SC-6: 20 [f][CIRC]

B3: 87.96 inches
Setup: (2 × π) × 14
SC-6: 2 [ENTER] [π] [×] 14 [×]

B4: 25 meters
Setup: $\sqrt{(15^2 + 20^2)}$
SC-6: 15 [ENTER] 20 [g][HYP]

B5: 20 inches
Setup: $\sqrt{(16^2 + 12^2)}$
SC-6: 16 [ENTER] 12 [g][HYP]

B6: 150.80 cm² (48π)
Setup: $1/3\pi(4^2)(9)$
SC-6: 4 [ENTER] 9 [g][CONE]

B7: 565.49 in² (180π)
Setup: $1/3\pi(6^2)(15)$
SC-6: 6 [ENTER] 15 [g][CONE]

B8: One 16-in pizza (201.06 in² > 157.08 in²)
Setup: $\pi \times 5^2 \times 2$
SC-6: 8 [f][CIRC] 5 [f][CIRC] 2 [×]

B9: 56.55 ft³
Setup: $\pi \times 6^2 \div 2$
SC-6: 6 [f][CIRC] 2 [÷]

B10: 17 feet
Setup: $\sqrt{(8^2 + 15^2)}$
SC-6: 8 [ENTER] 15 [g][HYP]

Unit 9: Statistical Measures & Data Displays — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + Probability Dice Studio (Probability Dice & Spinner)

Target Misconception: Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.

Hands-On Block Remediation: Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 9.1 (pp.86-87): STATISTICAL QUESTIONS, DOT PLOTS, AND THE MEAN

Methods: W1: Mean Formula (Sum ÷ Count) | W2: Fair-Share Block Leveling | W3: Balance Point Deviations

PRACTICE A (p.86): KEY IDEAS (#1-#10)

ExA: Sum = 105, Count = 5, Mean = 21

A1: Sum = 20, Mean = 5
Setup: Mean(2, 4, 6, 8)
SC-6: [g][CLΣ] 2 [Σ+] 4 [Σ+] 6 [Σ+] 8 [Σ+] [g][x-bar]

A2: Sum = 70, Mean = 14
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+] -> [g][x-bar]

A3: 156 cm
Setup: Mean(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][x-bar]

A4: Mean = 4.0 g
Setup: Load data -> Stats
SC-6: Load into [Σ+] -> [g][x-bar]

A5: Sum = 55, Mean = 18.33
Setup: Mean(35, 25)
SC-6: ... 35 [f][Σ-] 25 [Σ+] [g][x-bar]

A6: 13.33 (80 ÷ 6)
Setup: Mean(10, 4, 20, 2)
SC-6: 10 [ENTER] 4 [FRQ] 20 [ENTER] 2 [FRQ] [g][x-bar]

A7: 70°F
Setup: Sort/Load 6 values -> Stats
SC-6: Load 6 temps into [Σ+] -> [g][x-bar]

A8: 60 total blocks
Setup: 5 × 12
SC-6: 5 [ENTER] 12 [×]

A9: 95
Setup: 360 - 265
SC-6: 360 [ENTER] 265 [-]

A10: 17 points
Setup: (90 + 29) ÷ 7
SC-6: 90 [ENTER] 29 [+] 7 [÷]

PRACTICE B (p.87): REAL LIFE (#1-#10)

ExB: 94

B1: Sum = 85; Mean = 17 points
Setup: Mean(14, 18, 12, 22, 19)
SC-6: [g][CLΣ] 14 [Σ+] 18 [Σ+] 12 [Σ+] 22 [Σ+] 19 [Σ+] [g][x-bar]

B2: 42 grams
Setup: Mean(41, 43, 42, 40, 44...)
SC-6: [g][CLΣ] 41 [Σ+] 43 [Σ+] 42 [Σ+] 40 [Σ+] 44 [Σ+] 42 [Σ+] [g][x-bar]

B3: 89 ÷ 10 = 8.9
Setup: (((3 × 8) + (5 × 9)) + (2 × 10)) ÷ 10
SC-6: 3 [ENTER] 8 [×] 5 [ENTER] 9 [×] [+] 2 [ENTER] 10 [×] [+] 10 [÷]

B4: 91
Setup: (5 × 85) - (((80 + 88) + 82) + 84)
SC-6: 5 [ENTER] 85 [×] 80 [ENTER] 88 [+] 82 [+] 84 [+] [-]

B5: Total = 36,000; Mean = 9,000 steps
Setup: (((8200 + 9500) + 7800) + 10500) ÷ 4
SC-6: 8200 [ENTER] 9500 [+] 7800 [+] 10500 [+] 4 [÷]

B6: 82.5 -> 90.0 (+7.5 points)
Setup: ((88 + 92) + 90) ÷ 3
SC-6: 88 [ENTER] 92 [+] 90 [+] 3 [÷]

B7: 12 books per stack
Setup: (((6 + 10) + 14) + 18) ÷ 4
SC-6: 6 [ENTER] 10 [+] 14 [+] 18 [+] 4 [÷]

B8: 4.8 g
Setup: ((((4.2 + 5.1) + 4.8) + 5.5) + 4.4) ÷ 5
SC-6: 4.2 [ENTER] 5.1 [+] 4.8 [+] 5.5 [+] 4.4 [+] 5 [÷]

B9: Reading time varies from student to student
Setup: 60 × 1
SC-6: 60 [ENTER] 1 [×]

B10: 260 ÷ 10 = 26 points
Setup: (80 + 180) ÷ 10
SC-6: 80 [ENTER] 180 [+] 10 [÷]

SEC 9.2 (pp.88-89): MEASURES OF CENTER: MEDIAN, MODE, AND RANGE

Methods: W1: Ordered List (Least to Greatest) | W2: Dot Plot / Frequency Table Analysis

PRACTICE A (p.88): KEY IDEAS (#1-#10)

ExA: Mean = 7.875, Median = 7.5, Mode = 7, Range = 4

A1: Mode = 7
Setup: Sort/Load 8 values -> Mode
SC-6: Load 8 nums -> [g][MODE]

A2: Median = 14, Mode = 12, Range = 8
Setup: Sort/Load 6 values -> Med/Mode/Range
SC-6: Load 6 nums -> [f][MED], [g][MODE], [RNG]

A3: Mean = 9, Median = 8, Mode = 8, Range = 9
Setup: Sort/Load 5 values -> Stats
SC-6: Load 5 nums into [Σ+]

A4: Min = 62°F, Max = 85°F, Range = 23°F
Setup: Sort/Load 7 values -> Range
SC-6: Load 7 nums -> [f][MIN], [g][MAX], [RNG]

A5: 75°F
Setup: Sort/Load 6 values -> Med
SC-6: Load 6 nums -> [f][MED]

A6: Mean = $29.17, Median = $17.50 (Median is best)
Setup: Sort/Load 6 values -> Mean/Med
SC-6: Load 6 nums -> [g][x-bar], [f][MED]

A7: Mean increases from 14 to 30.8; Median stays 14
Setup: Compare both sets
SC-6: Mental / Table Check

A8: 95
Setup: 64 + 31
SC-6: 64 [ENTER] 31 [+]

A9: Median = 14, Range = 13
Setup: Sorted: 8, 11, 14, 16, 21
SC-6: Mental / Table Check

A10: Mean = $370k, Median = $220k (Use Median due to the $980k ou..
Setup: Sort/Load 5 values -> Mean/Med
SC-6: Load 5 nums -> [g][x-bar], [f][MED]

PRACTICE B (p.89): REAL LIFE (#1-#10)

ExB: Mean = $29.17, Median = $17.50, Mode = $15 (Median is best)

B1: Median = 62, Mode = 62, Range = 22
Setup: Med/Mode/Range(55, 62, 48, 70, 62...)
SC-6: [g][CLΣ] 55 [Σ+] 62 [Σ+] 48 [Σ+] 70 [Σ+] 62 [Σ+] 50 [Σ+] 66 [Σ+] [f][MED] [g][MODE] [..

B2: Median = 12.5, Mode = 12, Range = 5
Setup: Med(11, 13, 12, 14, 12...)
SC-6: [g][CLΣ] 11 [Σ+] 13 [Σ+] 12 [Σ+] 14 [Σ+] 12 [Σ+] 16 [Σ+] [f][MED]

B3: Mean=$40, Med=$25, Mode=$20; Median (outlier $105 skews mean)
Setup: Mean/Med(20, 25, 20, 30, 105)
SC-6: [g][CLΣ] 20 [Σ+] 25 [Σ+] 20 [Σ+] 30 [Σ+] 105 [Σ+] [g][x-bar] [f][MED]

B4: Modes = 6 and 8 (bimodal); Median = 7
Setup: 8
SC-6: 6 [ENTER] 8

B5: Median = 11.8 s, Range = 1.2 s, No mode
Setup: 12.4 - 11.2
SC-6: 12.4 [ENTER] 11.2 [-]

B6: Min = 64, Max = 81, Range = 17°F, Median = 72°F
Setup: Med/Range(72, 68, 75, 81, 64...)
SC-6: [g][CLΣ] 72 [Σ+] 68 [Σ+] 75 [Σ+] 81 [Σ+] 64 [Σ+] 70 [Σ+] 74 [Σ+] [f][MED] [RNG]

B7: Mean = 11, Median = 11, Mode = 12, Range = 7
Setup: Mean/Med/Mode/Range(8, 12, 9, 15, 12...)
SC-6: [g][CLΣ] 8 [Σ+] 12 [Σ+] 9 [Σ+] 15 [Σ+] 12 [Σ+] 10 [Σ+] [g][x-bar] [f][MED] [g][MODE] [R..

B8: Example: 8, 8, 10, 12, 15
Setup: 15 - 8
SC-6: 15 [ENTER] 8 [-]

B9: Median increases by 5 (to 85); Range stays 20
Setup: 95 - 75
SC-6: 95 [ENTER] 75 [-]

B10: Mean = 11.75, Median = 11.5, Mode = 10, Range = 6
Setup: Mean/Med(9, 12, 10, 14, 11...)
SC-6: [g][CLΣ] 9 [Σ+] 12 [Σ+] 10 [Σ+] 14 [Σ+] 11 [Σ+] 15 [Σ+] 10 [Σ+] 13 [Σ+] [g][x-bar] [f][M..

SEC 9.3 (pp.90-91): QUARTILES, INTERQUARTILE RANGE (IQR) & BOX PLOTS

Methods: W1: Split Ordered Data into Lower & Upper Halves | W2: Box-and-Whisker Plot & 25% Quartile Sections

PRACTICE A (p.90): KEY IDEAS (#1-#10)

ExA: Min = 68, Q1 = 73.5, Med = 82, Q3 = 90, Max = 96; IQR = 16.5

A1: Min = 10, Q1 = 15, Med = 30, Q3 = 45, Max = 50, IQR = 30
Setup: Sort/Load 5 values -> Q1/Q3/IQR
SC-6: Load 5 nums -> [f][1Q], [g][3Q], [IQR]

A2: Q1 = 6, Med = 10, Q3 = 14, IQR = 8
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [f][1Q], [g][3Q], [IQR]

A3: 34
Setup: 58 - 24
SC-6: 58 [ENTER] 24 [-]

A4: 75%
Setup: 3 quarters = 3 × 25%
SC-6: Mental / Table Check

A5: Q1 = 73.5, Med = 82, Q3 = 90, IQR = 16.5
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

A6: Q1 = 46.5, Med = 51, Q3 = 56.5, IQR = 10
Setup: Sort/Load 8 values -> Stats
SC-6: Load 8 nums into [Σ+]

A7: Player A (smaller IQR means the middle 50% of scores are clo..
Setup: Compare box widths
SC-6: Mental / Table Check

A8: 90 (Any score > 90 is an outlier)
Setup: 60 + (1.5 × 20)
SC-6: 60 [ENTER] 1.5 [ENTER] 20 [×] [+]

A9: Q1 = 15, Med = 22, Q3 = 28, IQR = 13
Setup: Sort/Load 7 values -> Q1/Q3/IQR
SC-6: Load 7 nums -> [IQR]

A10: About 100 students (50% of 200)
Setup: 50% of 200
SC-6: 200 [ENTER] 50 [%]

PRACTICE B (p.91): REAL LIFE (#1-#10)

ExB: IQR = 16; 20 students (50%)

B1: Min=40, Q1=45, Med=55, Q3=65, Max=70; IQR = 20
Setup: Q1/Q3/IQR(40, 45, 50, 55, 60...)
SC-6: [g][CLΣ] 40 [Σ+] 45 [Σ+] 50 [Σ+] 55 [Σ+] 60 [Σ+] 65 [Σ+] 70 [Σ+] [IQR]

B2: Q1 = 7.5, Med = 10, Q3 = 13, IQR = 5.5
Setup: Q1/Q3/IQR(6, 7, 8, 9, 11...)
SC-6: [g][CLΣ] 6 [Σ+] 7 [Σ+] 8 [Σ+] 9 [Σ+] 11 [Σ+] 12 [Σ+] 14 [Σ+] 15 [Σ+] [IQR]

B3: Q1 = 22.5, Med = 30, Q3 = 42.5, IQR = 20
Setup: Q1/Q3/IQR(30, 15, 45, 25, 35...)
SC-6: [g][CLΣ] 30 [Σ+] 15 [Σ+] 45 [Σ+] 25 [Σ+] 35 [Σ+] 20 [Σ+] 50 [Σ+] 40 [Σ+] 30 [Σ+] [IQR]

B4: Range = 28 ft; IQR = 13 ft
Setup: 31 - 18
SC-6: 40 [ENTER] 12 [-] 31 [ENTER] 18 [-]

B5: 75% = 45 trees
Setup: 75% of 60
SC-6: 60 [ENTER] 75 [%]

B6: 25% of 60 = 15 trees
Setup: 25% of 60
SC-6: 60 [ENTER] 25 [%]

B7: Class A (smaller IQR means less spread in the middle 50%)
Setup: 19 - 8
SC-6: 19 [ENTER] 8 [-]

B8: 30 + 15 = 45
Setup: 30 + (1.5 × 10)
SC-6: 30 [ENTER] 1.5 [ENTER] 10 [×] [+]

B9: Min=18, Q1=22, Med=27, Q3=34, Max=40; IQR = 12
Setup: Q1/Q3/IQR(18, 22, 25, 29, 34...)
SC-6: [g][CLΣ] 18 [Σ+] 22 [Σ+] 25 [Σ+] 29 [Σ+] 34 [Σ+] 40 [Σ+] [IQR]

B10: Range = 90, IQR = 8 (IQR is not distorted by the outlier 100)
Setup: Q1/Q3/IQR(10, 12, 14, 16, 18...)
SC-6: [g][CLΣ] 10 [Σ+] 12 [Σ+] 14 [Σ+] 16 [Σ+] 18 [Σ+] 20 [Σ+] 100 [Σ+] [IQR]

SEC 9.4 (pp.92-93): MEAN ABSOLUTE DEVIATION (MAD)

Methods: W1: 3-Step MAD Distance Table | W2: Number-Line Deviation Balance

PRACTICE A (p.92): KEY IDEAS (#1-#10)

ExA: Mean = 14, MAD = 2.4

A1: Mean = 156 cm, MAD = 4.0 cm
Setup: MAD(150, 162, 156)
SC-6: 150 [Σ+] 162 [Σ+] 156 [Σ+] [g][MAD]

A2: Mean = 14, MAD = 2.4
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A3: Mean = 20, MAD = 6.0
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A4: Mean = 7, MAD = 0 (Zero spread)
Setup: All distances from 7 are 0
SC-6: Mental / Table Check

A5: Mean = 28, MAD = 4.8
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A6: Mean = 13.0 s, MAD = 0.40 s
Setup: Sort/Load 6 values -> Mean/MAD
SC-6: Load 6 decimals -> [g][x-bar], [g][MAD]

A7: Mean = 88, MAD = 3.6
Setup: Sort/Load 5 values -> Mean/MAD
SC-6: Load 5 nums -> [g][x-bar], [g][MAD]

A8: Team B (a larger MAD indicates greater variation from the me..
Setup: Compare MADs
SC-6: Mental / Table Check

A9: Mean = 10; deviations: 4, 0, 1, 3; MAD = 2.0
Setup: Mean=10; dists 4+0+1+3=8 ÷ 4
SC-6: Mental / Table Check

A10: Mean = 9, Med = 8, Mode = 8, Range = 9, IQR = 5.5, MAD = 2.4
Setup: 1 entry -> 6 stats!
SC-6: Mental / Table Check

PRACTICE B (p.93): REAL LIFE (#1-#10)

ExB: Mean = 13.0 s; MAD = 0.40 s

B1: Mean = 210°C; MAD = 3.0°C
Setup: MAD(206, 208, 212, 214)
SC-6: [g][CLΣ] 206 [Σ+] 208 [Σ+] 212 [Σ+] 214 [Σ+] [g][MAD]

B2: Mean = 10; MAD = 1.6 shots
Setup: Mean/MAD(7, 9, 10, 11, 13)
SC-6: [g][CLΣ] 7 [Σ+] 9 [Σ+] 10 [Σ+] 11 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

B3: Mean = 50 g, MAD = 0 g (no deviation from the mean)
Setup: 0
SC-6: 0 [ENTER]

B4: Mean = 9 s, MAD = 2.0 s; Raven A is more consistent
Setup: MAD(6, 8, 10, 12)
SC-6: [g][CLΣ] 6 [Σ+] 8 [Σ+] 10 [Σ+] 12 [Σ+] [g][MAD]

B5: Mean = 85; MAD = 4.8
Setup: Mean/MAD(76, 82, 85, 88, 94)
SC-6: [g][CLΣ] 76 [Σ+] 82 [Σ+] 85 [Σ+] 88 [Σ+] 94 [Σ+] [g][x-bar] [g][MAD]

B6: Mean = 15 m, MAD = 1.0 m
Setup: Mean/MAD(15, 16, 14, 15, 17...)
SC-6: [g][CLΣ] 15 [Σ+] 16 [Σ+] 14 [Σ+] 15 [Σ+] 17 [Σ+] 13 [Σ+] [g][x-bar] [g][MAD]

B7: Mean = 85°F; deviations 5, 1, 1, 5; MAD = 3.0°F
Setup: MAD(80, 84, 86, 90)
SC-6: [g][CLΣ] 80 [Σ+] 84 [Σ+] 86 [Σ+] 90 [Σ+] [g][MAD]

B8: The signed deviations (-5 + 0 + 5) always add to 0
Setup: -5 + 5
SC-6: 5 [+/-] [ENTER] 5 [+]

B9: Mean = 5 min; MAD = 2.4 min
Setup: Mean/MAD(2, 3, 4, 5, 11)
SC-6: [g][CLΣ] 2 [Σ+] 3 [Σ+] 4 [Σ+] 5 [Σ+] 11 [Σ+] [g][x-bar] [g][MAD]

B10: Mean=16, Med=15, Mode=15, Rng=8, IQR=5.5, MAD=2.4
Setup: Mean/Med/Mode/Range/Q1/Q3/IQR/MAD(12, 15, 15, 18, 20)
SC-6: [g][CLΣ] 12 [Σ+] 15 [Σ+] 15 [Σ+] 18 [Σ+] 20 [Σ+] [g][x-bar] [f][MED] [g][MODE] [RNG] [..

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Practice A & B Complete Key

Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Hands-On Models: Ten-Frame Zero-Pair Counters (Ten-Frames & Two-Tone Counters) + Algebra Stage (Algebra & Area Tiles) + Probability Studio (Probability Dice & Spinner)

Target Misconception: Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).

Hands-On Block Remediation: Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x²] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 10.1 (pp.95-96): ADDING AND SUBTRACTING RATIONAL NUMBERS

Methods: W1: Zero-Pair Counters | W2: Number-Line Jumps | W3: Absolute Value Sign Rules & Add-the-Opposite

PRACTICE A (p.95): KEY IDEAS (#1-#10)

ExA: -8 + 5 = -3; -6 + (-4) = -10

A1: (a) -6; (b) -24
Setup: -15 + 9; -9
SC-6: 15 [+/-] [ENTER] 9 [+] / 9 [+/-] [+]

A2: -9
Setup: 14 + -23
SC-6: 14 [ENTER] 23 [+/-] [+]

A3: (a) 30; (b) -4
Setup: 12 - -18; -12
SC-6: 12 [ENTER] 18 [+/-] [-] / 12 [+/-]...

A4: -25
Setup: -9 - 16
SC-6: 9 [+/-] [ENTER] 16 [-]

A5: -14.2°C
Setup: (0 - 7.8) - 6.4
SC-6: 0 [ENTER] 7.8 [-] 6.4 [-]

A6: -4.8
Setup: 6.4 - 11.2
SC-6: 6.4 [ENTER] 11.2 [-]

A7: +2.7
Setup: -7.5 - -10.2
SC-6: 7.5 [+/-] [ENTER] 10.2 [+/-] [-]

A8: -1/8
Setup: -3/4 - -5/8
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-]

A9: 36°F
Setup: 14 - -22
SC-6: 14 [ENTER] 22 [+/-] [-]

A10: 2 7/8 (2.875)
Setup: (-2.5 + 4.25) + 1.125
SC-6: 2.5 [+/-] [ENTER] 4.25 [+] 1.125 [+]

PRACTICE B (p.96): REAL LIFE (#1-#10)

ExB: 14°C; and -1/8

B1: +8°F
Setup: -14 + 22
SC-6: 14 [+/-] [ENTER] 22 [+]

B2: -83 meters
Setup: -45 + -38
SC-6: 45 [+/-] [ENTER] 38 [+/-] [+]

B3: 4,800 - (-200) = 5,000 meters
Setup: 4800 - -200
SC-6: 4800 [ENTER] 200 [+/-] [-]

B4: -15 - (-25) = +10 points
Setup: -15 - -25
SC-6: 15 [+/-] [ENTER] 25 [+/-] [-]

B5: +$8.75
Setup: (-12.50 + 30) - 8.75
SC-6: 12.50 [+/-] [ENTER] 30 [+] 8.75 [-]

B6: 1/8
Setup: -5/8 + 3/4
SC-6: [.] 5 [.] 8 [+/-] [ENTER] [.] 3 [.] 4 [+]

B7: -9/6 = -1 1/2
Setup: -2/3 - 5/6
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 5 [.] 6 [-]

B8: -16.0 meters
Setup: (-18.4 + 7.9) - 5.5
SC-6: 18.4 [+/-] [ENTER] 7.9 [+] 5.5 [-]

B9: -15.0°C
Setup: -8.5 - 6.5
SC-6: 8.5 [+/-] [ENTER] 6.5 [-]

B10: -8
Setup: ((-14 - -20) + -9) - 5
SC-6: 14 [+/-] [ENTER] 20 [+/-] [-] 9 [+/-] [+] 5 [-]

SEC 10.2 (pp.97-98): MULTIPLYING AND DIVIDING RATIONAL NUMBERS

Methods: W1: Same/Different Sign Rules | W2: Repeated Addition / Equal Number-Line Jumps & Counter Flips

PRACTICE A (p.97): KEY IDEAS (#1-#10)

ExA: 6×(-4) = -24, (-6)×(-4) = +24, $(-5)^3$ = +25

A1: (a) -56; (b) +56
Setup: 7 × -8; -7
SC-6: 7 [ENTER] 8 [+/-] [×] / 7 [+/-]...

A2: (a) -5; (b) +5
Setup: -45 ÷ 9; -9
SC-6: 45 [+/-] [ENTER] 9 [÷] / 9 [+/-] [÷]

A3: +25 (2 negative factors = positive)
Setup: -5^2
SC-6: 5 [+/-] [x²]

A4: -8 (3 negative factors = odd count = negative)
Setup: -2^3
SC-6: 2 [+/-] [x³]

A5: +8.4
Setup: -3.5 × -2.4
SC-6: 3.5 [+/-] [ENTER] 2.4 [+/-] [×]

A6: -12
Setup: -14.4 ÷ 1.2
SC-6: 14.4 [+/-] [ENTER] 1.2 [÷]

A7: -3/5 (-0.6)
Setup: -2/3 × 9/10
SC-6: [.] 2 [.] 3 [+/-] [ENTER] [.] 9 [.] 10 [×]

A8: +2
Setup: -5/6 ÷ -5/12
SC-6: [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷]

A9: -120
Setup: (-4 × -5) × -6
SC-6: 4 [+/-] [ENTER] 5 [+/-] [×] 6 [+/-] [×]

A10: -76
Setup: (-18 × 4) - (-36 ÷ -9)
SC-6: 18 [+/-] [ENTER] 4 [×] 36 [+/-] [ENTER] 9 [+/-] [÷] [-]

PRACTICE B (p.98): REAL LIFE (#1-#10)

ExB: 2 minutes

B1: -120 feet
Setup: 8 × -15
SC-6: 8 [ENTER] 15 [+/-] [×]

B2: +40 points (same signs give a positive product)
Setup: -4 × -10
SC-6: 4 [+/-] [ENTER] 10 [+/-] [×]

B3: -4.5°F per hour
Setup: -27 ÷ 6
SC-6: 27 [+/-] [ENTER] 6 [÷]

B4: $(-4)^3$ = +16; $(-4)^3$ = -64
Setup: -4^6
SC-6: 4 [+/-] [x³] 4 [+/-] [x³]

B5: +2/3
Setup: -3/4 × -8/9
SC-6: [.] 3 [.] 4 [+/-] [ENTER] [.] 8 [.] 9 [+/-] [×]

B6: -2
Setup: -7/8 ÷ 7/16
SC-6: [.] 7 [.] 8 [+/-] [ENTER] [.] 7 [.] 16 [÷]

B7: -12
Setup: (-2.5 × -1.6) × -3
SC-6: 2.5 [+/-] [ENTER] 1.6 [+/-] [×] 3 [+/-] [×]

B8: -3.2 meters per year
Setup: -14.4 ÷ 4.5
SC-6: 14.4 [+/-] [ENTER] 4.5 [÷]

B9: +120
Setup: ((-2 × -3) × -4) × -5
SC-6: 2 [+/-] [ENTER] 3 [+/-] [×] 4 [+/-] [×] 5 [+/-] [×]

B10: 36 - (-20) + 6 = 62
Setup: $(-6^2 - (4 \times -5)) + (-18 \div -3)$
SC-6: 6 [+/-] [x²] 4 [ENTER] 5 [+/-] [×] [-] 18 [+/-] [ENTER] 3 [+/-..

SEC 10.3 (pp.99-100): SOLVING TWO-STEP EQUATIONS (AX + B = C)

Methods: W1: Two-Step Inverse Operations | W2: Balance Scale / Bar Model Unwrapping

PRACTICE A (p.99): KEY IDEAS (#1-#10)

ExA: x = 7 (Step 1: 3x = 21)

A1: 4x = 36; x = 9
Setup: (45 - 9) ÷ 4
SC-6: 45 [ENTER] 9 [-] 4 [÷]

A2: 5x = 50; x = 10
Setup: (38 + 12) ÷ 5
SC-6: 38 [ENTER] 12 [+] 5 [÷]

A3: 6y = 48; y = 8
Setup: (59 - 11) ÷ 6
SC-6: 59 [ENTER] 11 [-] 6 [÷]

A4: 8n = 64; n = 8
Setup: (49 + 15) ÷ 8
SC-6: 49 [ENTER] 15 [+] 8 [÷]

A5: m / 4 = 9; m = 36
Setup: (15 - 6) × 4
SC-6: 15 [ENTER] 6 [-] 4 [×]

A6: k / 3 = 12; k = 36
Setup: (7 + 5) × 3
SC-6: 7 [ENTER] 5 [+] 3 [×]

A7: -3x = -12; x = 4
Setup: (2 - 14) ÷ -3
SC-6: 2 [ENTER] 14 [-] 3 [÷/-] [÷]

A8: 2.5w = 15; w = 6
Setup: (23 - 8) ÷ 2.5
SC-6: 23 [ENTER] 8 [-] 2.5 [÷]

A9: 25h + 35 = 160; h = 5 hours
Setup: (160 - 35) ÷ 25
SC-6: 160 [ENTER] 35 [-] 25 [÷]

A10: 15w + 45 = 120; w = 5 weeks
Setup: (120 - 45) ÷ 15
SC-6: 120 [ENTER] 45 [-] 15 [÷]

PRACTICE B (p.100): REAL LIFE (#1-#10)

ExB: 4 hours

B1: 6r + 8 = 38 -> r = 5 laps
Setup: (38 - 8) ÷ 6
SC-6: 38 [ENTER] 8 [-] 6 [÷]

B2: 4h + 12 = 48 -> h = 9 hours
Setup: (48 - 12) ÷ 4
SC-6: 48 [ENTER] 12 [-] 4 [÷]

B3: 15w + 35 = 140 -> w = 7 weeks
Setup: (140 - 35) ÷ 15
SC-6: 140 [ENTER] 35 [-] 15 [÷]

B4: 4b - 10 = 46 -> b = $14 per book
Setup: (46 + 10) ÷ 4
SC-6: 46 [ENTER] 10 [+] 4 [÷]

B5: x = 6 (-4·6 + 18 = -6)
Setup: (-6 - 18) ÷ -4
SC-6: 6 [+/-] [ENTER] 18 [-] 4 [+/-] [÷]

B6: n = 35
Setup: (16 - 9) × 5
SC-6: 16 [ENTER] 9 [-] 5 [×]

B7: 3000 - 40m = 600 -> m = 60 minutes
Setup: (600 - 3000) ÷ -40
SC-6: 600 [ENTER] 3000 [-] 40 [+/-] [÷]

B8: x = 24
Setup: (11 + 5) ÷ 2/3
SC-6: 11 [ENTER] 5 [÷] [.] 2 [.] 3 [÷]

B9: 2w + 36 = 56 -> w = 10 cm
Setup: (56 - 36) ÷ 2
SC-6: 56 [ENTER] 36 [-] 2 [÷]

B10: 2.25m + 3.50 = 21.50 -> m = 8 miles
Setup: (21.50 - 3.50) ÷ 2.25
SC-6: 21.50 [ENTER] 3.50 [-] 2.25 [÷]

SEC 10.4 (pp.101-102): THEORETICAL PROBABILITY, COMBINATIONS & PERMUTATIONS

Methods: W1: Probability Ratio & Complement Rule | W2: Tree Diagram / Outcome List | W3: Fundamental Counting Principle

PRACTICE A (p.101): KEY IDEAS (#1-#10)

ExA: P(Blue) = 2/5 = 0.4 = 40%; 5! = 120 ways

A1: P(5) = 1/6; P(not 6) = 5/6
Setup: 1/6; and 1 - 1/6 = 5/6
SC-6: Mental / Table Check

A2: 0.65 (65% or 13/20)
Setup: 1 - 0.35
SC-6: 1 [ENTER] 0.35 [-]

A3: P(double 6s) = 1/36; P(sum 11 or 12) = 3/36 = 1/12
Setup: 1/36; and 3/36 = 1/12
SC-6: Mental / Table Check

A4: 24 ways
Setup: 4!
SC-6: 4 [x!]

A5: 720 ways
Setup: 6!
SC-6: 6 [x!]

A6: 56 teams
Setup: C(8,3)
SC-6: 8 [ENTER] 3 [f] [nCr]

A7: 336 podium arrangements
Setup: P(8,3)
SC-6: 8 [ENTER] 3 [g] [nPr]

A8: 1/8 (12.5%)
Setup: 1/2 × 1/2 × 1/2 = 1/8
SC-6: Mental / Table Check

A9: 1/3 (6/10 × 5/9 = 30/90)
Setup: 6/10 × 5/9 = 30/90
SC-6: Mental / Table Check

A10: 120 combinations vs. 720 permutations (differ by 3! = 6)
Setup: C(10,3); gnPr
SC-6: 10 [ENTER] 3 [f] [nCr] / [g] [nPr]

PRACTICE B (p.102): REAL LIFE (#1-#10)

ExB: 720 ordered podiums; 120 unordered 3-runner crews

B1: 4/20 = 1/5 = 0.20 = 20%
Setup: Simp(4/20)
SC-6: [.] 4 [.] 20 [g] [SIMP] [FDISP]

B2: 65% (13/20 = 0.65)
Setup: 100 - 35
SC-6: 100 [ENTER] 35 [-]

B3: P(black) = 2/5; P(not white) = 11/15
Setup: Simp(6/15)
SC-6: [.] 6 [.] 15 [g] [SIMP]

B4: 120 custom skateboards
Setup: (5 × 4) × 6
SC-6: 5 [ENTER] 4 [×] 6 [×]

B5: 5! = 120 orders
Setup: 5!
SC-6: 5 [x!]

B6: 210 batting orders (permutations)
Setup: P(7,3)
SC-6: 7 [ENTER] 3 [g] [nPr]

B7: 35 lab groups (combinations)
Setup: C(7,3)
SC-6: 7 [ENTER] 3 [f] [nCr]

B8: 28 fruit combinations
Setup: C(8,2)
SC-6: 8 [ENTER] 2 [f] [nCr]

B9: 60 times
Setup: 160 × 3/8
SC-6: 160 [ENTER] [.] 3 [.] 8 [×]

B10: 12 outcomes; P(H and 6) = 1/12
Setup: 2 × 6
SC-6: 2 [ENTER] 6 [×]

Middle School 10-Unit Misconception & Hands-On Remediation Matrix

UNIT & HANDS-ON MODEL	TARGET CLASSROOM MISCONCEPTION	HANDS-ON BLOCK & SC-6 REMEDIATION
Unit 1: Numerical Expressi Counting & Array Studio	Students interpret remainders as meaningless decimal tails (e.g., .875 of a person), read 2^3 as 2×3=6, lose track of PEMDAS parentheses, and confuse GCF (overlap) with LCM (union).	Have students build physical tile rows for division remainders, stack 2D/3D arrays for exponents, use the 4-line elevator (Y holds Room 1 while X solves Room 2) instead of rewriting PEMDAS lines, and stand logarithmic Prime Factor Towers side-by-side on the 12-socket lid.
Unit 2: Fractions & Decima Fraction Studio (Fractio	Students treat fractions, mixed numbers, and decimals as unrelated rules, expect multiplication to always make numbers bigger, forget which fraction to flip in division, and misalign decimal columns.	Use the 84.0 mm Fraction Studio towers so equivalence and common denominators are physical millimeter height matches. Use [f] [1/x] on Floor 1 (X) so students see only the divisor flip, and use [f] [IP] / [g] [FP] to split whole dollars from decimal cents.
Unit 3: Ratios, Rates & Pe Counting & Array Tiles (	Students confuse Part-to-Part (3:2) with Part-to-Whole (3/5), add instead of multiply in ratio tables, divide unit rates upside-down, and memorize 6 disconnected rules for moving decimal points in percents.	Use Blue and Gold Counting Tiles as physical Tape Diagram boxes, step through ratio tables rhythmically with [g] [REP], use [x swap y] when unit-rate division is entered upside-down, and use [f] [%] + [FDISP] to show Fraction/Decimal/Percent are three costumes of one number.
Unit 4: Algebraic Expressi Algebra Boxes & Factorin	Students panic when letters appear, erroneously combine $3x + 2$ into $5x$ or $x^2 + x$ into x^3, and don't see why $3(x + 4) = 3x + 12$.	Show that a variable x is literally a physical green x-rod (42×14mm) whereas 1 is a small yellow unit cube (14×14mm) and x^2 is a large blue flat (42×42mm)—different physical shapes cannot be counted as one pile! Build rectangles on the L-frame stage for the Distributive Property.
Unit 5: Areas of Polygons Counting & Array Tiles (	Students multiply the slanted side of a parallelogram instead of the vertical height, forget to divide by 2 for triangles, lose track of multi-piece composite areas, and subtract clock times in base-10 instead of base-60.	Slide the corner triangle of a parallelogram to form a straight [f] [RECT] rectangle, use the dedicated [g] [TRI] key ([g][7]) which divides by 2 automatically, accumulate composite chunks with [Σ+], and use the Planetary Clock + [CLK] key for base-60 time.
Unit 6: Integers, Rational Ten-Frame Two-Tone Count	Students confuse the subtraction operator [-] with the negative sign [+/-], think $-18 > -5$ because $18 > 5$, confuse Opposite ([+/-]) with Absolute Value ([g] [|x|]), and reverse (x, y) coordinates.	Use two-tone (+1 / -1) Zero-Pair counters and Zero-Pair tiles, plot signed jumps on the SC-6 Number Line ([f] [LINE]), test inequalities with [f] [>] and [g] [<], contrast [+/-] (flip sign) with [g] [|x|] (always positive distance from 0), and use the Elevator Rule (walk across x first, ride up/down y second).
Unit 7: Equations & Inequa Algebra Boxes & Stage (A	Students treat '=' as 'write the answer', translate '8 less than n' backwards as 8 - n, resent writing -14 under both sides of $x + 14 = 39$, and confuse open vs. solid inequality circles.	Model '=' as a Balance Scale using Algebra Boxes & Stage. Solve 1-step equations with inverse operations on both sides, verify your solution with the SC-6 Equality Test [!=] (which flashes NO when Left = Right!), and test inequalities on [f] [LINE] with [f] [>=] and [g] [<=] (which flash YES or NO).
Unit 8: Surface Area & Vol Counting & Array Tiles (	Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).	Have students touch the 3 pairs of faces on their physical box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].
Unit 9: Statistical Measur Counting & Array Tiles (	Students forget to sort data before finding Median/Quartiles, lose their place adding long lists for Mean, and dread Mean Absolute Deviation (MAD) because doing it by hand takes 20 minutes of subtraction.	Show that Mean is physically leveling unequal cube towers to the same height. Then show that entering a list ONCE with [Σ+] unlocks Mean ([g] [x-bar]), Median ([f] [MED]), Mode ([g] [MODE]), Range ([RNG]), Quartiles ([f] [1Q], [g] [3Q], [IQR]), and MAD ([g] [MAD]) in 1 click each!
Unit 10: Grade 7 Bridge — S Ten-Frame Zero-Pair Coun	Students guess signs when adding/subtracting negatives, get -25 instead of +25 for $(-5)^2$ on algebraic calculators, undo ×3 before +7 in $3x + 7 = 28$, and confuse Permutations (order matters) with Combinations (order doesn't).	Use Yellow/Red counters to show removing 4 red debt tokens (10 - (-4)) equals +14. Notice how RPN 5 [+/-] [x^2] squares the entire register $(-5)^2 = +25$ without parentheses! Unwrap 2-step equations in reverse (untie ribbon ±b first, open box ×a second), and finish with [DICE], [f] [nCr], [g] [nPr], and [f] [?].

Photocopiable 10-Unit Middle School Student Mastery Tracker

STUDENT NAME / PERIOD	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
01. _____										
02. _____										
03. _____										
04. _____										
05. _____										
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16. _____										
17. _____										
18. _____										
19. _____										
20. _____										

TEACHER OBSERVATION NOTES & INTERVENTION LOG

- **Spiral-Bound Student Workbooks & Teacher Guides (Lulu): https://www.lulu.com/spotlight/stackcalc**
- Companion Book — From Blocks to the Stack (Amazon): https://link.amazon/B0f5cP0Ur | Manipulatives: https://stackcalc.io